

How Quickly Does a Tiny Neural Prefetcher Become Useful?

Causal Online Learning for 602.gcc_s-734B

602 Stride online experiment

Measured simulator/CPU co-simulation

1 Measured answer and scope

This report contains 14 completed primary cold-start runs and 10 completed compatibility or finite-resource runs. The analysis discovered 24 run directories; 24 had a finished simulator log with the required raw counters. Software fixtures and development pilots are excluded from research comparisons.

Among 8 completed functional online arms, 8 saved total measured cycles relative to the matched conventional Stride run. These are per-run observations on one trace slice, not statistical equivalence or a hardware implementation claim.

For h8 random-start learning (completed seeds 7, 17, 27), 3/3 runs recovered startup cost versus Stride. The first qualifying endpoint was 1.0M retired instructions from the slice boundary, confirmed at 3.0M. At first attainment the learner had observed 16,196 eligible callbacks and completed 253 updates.

For h8 random-start learning (completed seeds 7, 17, 27), 3/3 runs attained 99% of same-H frozen-ilm window IPC. The first qualifying endpoint was 2.0M retired instructions from the slice boundary, confirmed at 4.0M. At first attainment the learner had observed 32,320 eligible callbacks and completed 505 updates.

Over all 25M measured instructions, these runs saved 6.860–6.912 million cycles versus Stride. Their useful-prefetch coverage was 57.10–57.82%, selected accuracy 26.23–26.50%, and request pressure 1.086–1.111 requests per L2 LOAD. These dimensions are reported separately; close IPC does not imply equal action behavior.

This is 10.02–10.09% fewer cycles with 3.63–3.71 times Stride’s requested traffic. The matched Stride coverage/selected accuracy are 17.13%/28.84%. The measured trade-off is useful misses covered versus request pressure; lower precision alone is not a cache-pollution measurement.

Their end-to-end IPC is 99.884–99.968% of the matched same-H frozen-ilm IPC. This end-to-end ratio includes startup and is separate from the three-window attainment milestone.

For h8 ilm warm-start adaptation (completed seeds 7), 1/1 runs recovered startup cost versus Stride. The first qualifying endpoint was 1.0M retired instructions from the slice boundary, confirmed at 3.0M. At first attainment the learner had observed 16,196 eligible callbacks and completed 253 updates.

For h8 ilm warm-start adaptation (completed seeds 7), 1/1 runs attained 99% of same-H frozen-ilm window IPC. The first qualifying endpoint was 1.0M retired instructions from the slice boundary, confirmed at 3.0M. At first attainment the learner had observed 16,196 eligible callbacks and completed 253 updates.

Over all 25M measured instructions, these runs saved 6.896 million cycles versus Stride. Their useful-prefetch coverage was 56.92%, selected accuracy 26.76%, and request pressure 1.072 requests per L2 LOAD. These dimensions are reported separately; close IPC does not imply equal action behavior.

This is 10.07% fewer cycles with 3.58 times Stride’s requested traffic. The matched Stride coverage/selected accuracy are 17.13%/28.84%. The measured trade-off is useful misses covered versus request pressure; lower precision alone is not a cache-pollution measurement.

Their end-to-end IPC is 99.942% of the matched same-H frozen-ilm IPC. This end-to-end ratio includes startup and is separate from the three-window attainment milestone.

For h16 random-start learning (completed seeds 7, 17, 27), 3/3 runs recovered startup cost versus Stride. The first qualifying endpoint was 1.0M retired instructions from the slice boundary, confirmed at 3.0M. At first attainment the learner had observed 16,196 eligible callbacks and completed 253 updates.

For h16 random-start learning (completed seeds 7, 17, 27), 3/3 runs attained 99% of same-H frozen-ilm window IPC. The first qualifying endpoint was 2.0M retired instructions from the slice boundary, confirmed at 4.0M. At first attainment the learner had observed 32,320 eligible callbacks and completed 505 updates.

Over all 25M measured instructions, these runs saved 5.280–6.519 million cycles versus Stride. Their useful-prefetch coverage was 47.14–54.13%, selected accuracy 26.44–26.63%, and request pressure 0.894–1.028 requests per L2 LOAD. These dimensions are reported separately; close IPC does not imply equal action behavior.

This is 7.71–9.52% fewer cycles with 2.99–3.43 times Stride’s requested traffic. The matched Stride coverage/selected accuracy are 17.13%/28.84%. The measured trade-off is useful misses covered versus request pressure; lower precision alone is not a cache-pollution measurement.

Their end-to-end IPC is 97.278–99.223% of the matched same-H frozen-ilm IPC. This end-to-end ratio includes startup and is separate from the three-window attainment milestone.

The h16 random-start learning result is not permanent frozen-model attainment: in the final 1M-instruction window, IPC was 96.32–96.41% of same-H frozen-ilm, coverage was 20.34–20.69%, and request pressure had fallen to 0.379–0.400. The curves show late regression after the initial qualifying three-window streak. With unbounded state and zero modeled service delay, this is not a finite-table or engine-congestion loss. It accompanies a changing learned action policy while optimization continues; the imitation objective does not optimize cache usefulness or IPC. The completed measurements are retained without tuning the recipe to remove this unsuccessful aspect of the capacity check.

For h16 ilm warm-start adaptation (completed seeds 7), 1/1 runs recovered startup cost versus Stride. The first qualifying endpoint was 1.0M retired instructions from the slice boundary, confirmed at 3.0M. At first attainment the learner had observed 16,196 eligible callbacks and completed 253 updates.

For h16 ilm warm-start adaptation (completed seeds 7), 1/1 runs attained 99% of same-H frozen-ilm window IPC. The first qualifying endpoint was 1.0M retired instructions from the slice boundary, confirmed at 3.0M. At first attainment the learner had observed 16,196 eligible callbacks and completed 253 updates.

Over all 25M measured instructions, these runs saved 4.421 million cycles versus Stride. Their useful-prefetch coverage was 42.12%, selected accuracy 26.74%, and request pressure 0.794 requests per L2 LOAD. These dimensions are reported separately; close IPC does not imply equal action behavior.

This is 6.45% fewer cycles with 2.65 times Stride’s requested traffic. The matched Stride coverage/selected accuracy are 17.13%/28.84%. The measured trade-off is useful misses covered versus request pressure; lower precision alone is not a cache-pollution measurement.

Their end-to-end IPC is 95.974% of the matched same-H frozen-ilm IPC. This end-to-end ratio includes startup and is separate from the three-window attainment milestone.

The h16 ilm warm-start adaptation result is not permanent frozen-model attainment: in the final 1M-instruction window, IPC was 97.05% of same-H frozen-ilm, coverage was 20.01%, and request pressure had fallen to 0.372. The curves show late regression after the initial qualifying three-window streak. With unbounded state and zero modeled service delay, this is not a finite-table or engine-congestion loss. It accompanies a changing learned action policy while optimization continues; the imitation objective does not optimize cache usefulness or IPC. The completed measurements are retained without tuning the recipe to remove this unsuccessful aspect of the capacity check.

With the same 64-entry table and the predeclared 4-MAC/cycle service model, 0/2 completed frozen/scratch arms beat Stride end to end. Their IPC was 0.362523–0.362539, useful coverage 0.239–0.258%, and only 20.71–20.80% of eligible NN decisions completed. The scratch learner completed 253 updates, with 320,514 inference input drops and 67,420 training admission drops. The algorithmic zero-cost usefulness therefore does not establish a timing-costed hardware win. Frozen and online service measurements use identical queue/table limits; the conventional Stride comparator retains its original callback-time requests.

With the same 64-entry table and the predeclared 16-MAC/cycle service model, 0/2 completed frozen/scratch arms beat Stride end to end. Their IPC was 0.362708–0.362900, useful coverage 0.611–0.643%, and only 37.61–37.92% of eligible NN decisions completed. The scratch learner completed 440 updates, with 250,945 inference input drops and 125,021 training admission drops. The algorithmic zero-cost usefulness therefore does not establish a timing-costed hardware win. Frozen and online service measurements use identical queue/table limits; the conventional Stride comparator retains its original callback-time requests.

2 From offline imitation to online updates

The summer experiment trained the compact PC-keyed LSTM from Stride supervision and replayed its direct signed-delta actions. The next experiment moved the same frozen model into C++ live ChampSim inference: weights remained fixed while recurrent state changed. The present experiment adds progressive weight updates from labels of already observed events. Frozen inference, random-start learning, and ilm warm-start adaptation are different conditions. The ilm warm start includes offline training and is never learning from zero observations.

PC and aligned cache-line address are the only neural prediction features. The learned hurdle determines whether to emit, the learned positive count determines K , and the GRU decoder generates direct signed deltas with free-running feedback. Teacher actions supervise loss only. The shadow Stride does not issue prefetches in an NN arm. Its current-line first action is preserved, rather than changing Stride to a different textbook policy.

2.1 Causality and recipe

Each eligible demand is predicted with available recurrent state and a coherent weight version before its label may enter training. Batches are chronological; same-PC grouping is limited to already observed events inside a batch, using saved detached starting states. State lifetimes distinguish eviction and reallocation. Publication carries detached h/c forward. A fresh optimizer belongs to each independent run. The teacher and outcomes never enter the feature encoder.

The locked recipe takes one Adam step at learning rate 0.002 per 64 newly admitted chronological decisions, one pass per batch, with one training-library thread. The authoritative hurdle/count/direct-delta loss and free-running decoder recurrence are reused. Hurdle weights remain (1, 1) until both classes have appeared in training-available labels; thereafter they are $(N/(2N_-), N/(2N_+))$ using only cumulative received labels, including the current completed batch. Labels dropped before training admission do not supply balancing statistics. Empty, single-class and short batches are valid; the final incomplete pending batch is not flushed after the simulation boundary. TBPTT spans the actual same-lifetime subsequence inside a batch, at most 64 steps, not 256 same-PC steps or 64 steps for every PC.

2.2 Software validation

Correctness fixtures cover frozen Python/C++ parity, future-prefix independence, real gradients and parameter changes, single-class/empty/short batches, state eviction lifetimes, delayed completion without another demand callback, coherent publication, occupancy accounting, and timely/late/unused/unresolved lifecycle outcomes. These checks validate software behavior and are not research measurements. A saved-activation measurement hook was corrected to retain detached aliases instead of gradient-history cycles; a controlled 500-update check produced byte-identical final weights. Final online measurements were rerun with the corrected hook and unchanged learning recipe.

Locked recipe field	Value
jobs	2
batch_decisions	64
learning_rate	0.002
state_capacity	64
input_queue	16
output_queue	32
decoder_limit	32
nonlinear_cycles	4
development_skip	20000000
pilot_instructions	100000
final_skip	25000000
final_instructions	25000000

2.3 Two starting conditions

Historical compatibility uses 25M no-prefetch cache warmup plus 25M measurement and empty parity predictor state at the boundary. Its NN service cost is zero and state routing unbounded. The conventional live Stride reference from earlier work trained and issued during warmup; its lifecycle counts have a different scope and cannot replace the matched comparator.

The primary protocol is *cold start at the held-out trace-slice boundary*, corresponding to the original 25M–50M instruction region. Skipped records are not simulated and do not warm caches. Frozen arms load only trained weights; caches, recurrent histories, teacher, and online optimizer otherwise begin fresh. This is not program startup. All cold arms execute the same selected target instruction records; callback order and timing may change with the closed-loop prefetcher.

Retired target instructions from the slice origin are the horizontal axis. Eligible L2 callbacks, admitted/completed decisions, available labels/actions, updates, training exposures, and published versions are separate counters. Read-ahead trace records do not count as predictor observations. The original retirement-group warmup actually ends after 25,000,004 retired instructions, and compatibility measurement retires 25,000,003 instructions. These small overshoots are preserved and recorded separately from the nominal 25M protocol. Cold runs skip exactly 25,000,000 records and retire exactly 25,000,000 selected instructions.

3 End-to-end measurements

IPC and cycles include startup. Prefetch quality below uses legacy counter meanings and matched no-prefetch denominators. Raw counts are retained in the committed CSV tables. NA denotes an undefined ratio or missing observation.

Historical compatibility coverage and miss reduction reuse the completed original same-protocol no-prefetch *end counters*: 25,000,003 measured instructions and 203,131 L2 LOAD misses. This is a reference, not another new run. It is never used for cold-start comparisons or dynamic windows. The earlier conventional live Stride warmup reference is not reused for cycle-saving comparisons.

Run	Protocol	Origin records	Warm M	Weights	Seed	State	MACs/cycle
h16 frozen ilm	cold	25,000,000	0	ilm	7	unbounded	0
h8 frozen ilm	cold	25,000,000	0	ilm	7	unbounded	0
h8 frozen ilm /64 /MAC0	cold	25,000,000	0	ilm	7	64	0
h8 frozen ilm /64 /MAC16	cold	25,000,000	0	ilm	7	64	16
h8 frozen ilm /64 /MAC4	cold	25,000,000	0	ilm	7	64	4
h16 frozen i20m	cold	25,000,000	0	i20m	7	unbounded	0
h8 frozen i20m	cold	25,000,000	0	i20m	7	unbounded	0
No prefetch	cold	25,000,000	0	NA	7	NA	0
h16 scratch s17	cold	25,000,000	0	random	17	unbounded	0
h16 scratch s27	cold	25,000,000	0	random	27	unbounded	0
h16 scratch s7	cold	25,000,000	0	random	7	unbounded	0
h8 scratch s17	cold	25,000,000	0	random	17	unbounded	0
h8 scratch s27	cold	25,000,000	0	random	27	unbounded	0
h8 scratch s7	cold	25,000,000	0	random	7	unbounded	0
h8 scratch s7 /64 /MAC0	cold	25,000,000	0	random	7	64	0
h8 scratch s7 /64 /MAC16	cold	25,000,000	0	random	7	64	16
h8 scratch s7 /64 /MAC4	cold	25,000,000	0	random	7	64	4
h16 warm ilm s7	cold	25,000,000	0	ilm	7	unbounded	0
h8 warm ilm s7	cold	25,000,000	0	ilm	7	unbounded	0
Stride	cold	25,000,000	0	NA	7	64 Stride	0
h16 frozen ilm	historical	25,000,004	25	ilm	7	unbounded	0
h8 frozen ilm	historical	25,000,004	25	ilm	7	unbounded	0
h16 frozen i20m	historical	25,000,004	25	i20m	7	unbounded	0
h8 frozen i20m	historical	25,000,004	25	i20m	7	unbounded	0

Run	IPC	Cycles (M)	Saved vs Stride (M)	L2 MPKI
h16 frozen ilm	0.40652	61.497	7.001	3.401
h8 frozen ilm	0.40607	61.566	6.931	3.419
h8 frozen ilm /64 /MAC0	0.40606	61.567	6.931	3.419
h8 frozen ilm /64 /MAC16	0.36290	68.889	-0.392	8.098
h8 frozen ilm /64 /MAC4	0.36254	68.958	-0.460	8.130
h16 frozen i20m	0.40682	61.452	7.045	3.391
h8 frozen i20m	0.40625	61.538	6.959	3.444
No prefetch	0.36251	68.964	-0.466	8.151
h16 scratch s17	0.39949	62.580	5.917	4.015
h16 scratch s27	0.39546	63.218	5.280	4.309
h16 scratch s7	0.40336	61.979	6.519	3.739

Run	IPC	Cycles (M)	Saved vs Stride (M)	L2 MPKI
h8 scratch s17	0.40578	61.610	6.888	3.470
h8 scratch s27	0.40594	61.586	6.912	3.438
h8 scratch s7	0.40560	61.638	6.860	3.497
h8 scratch s7 /64 /MAC0	0.40560	61.637	6.860	3.495
h8 scratch s7 /64 /MAC16	0.36271	68.926	-0.428	8.101
h8 scratch s7 /64 /MAC4	0.36252	68.961	-0.463	8.131
h16 warm ilm s7	0.39016	64.077	4.421	4.718
h8 warm ilm s7	0.40583	61.602	6.896	3.511
Stride	0.36498	68.498	0.000	6.755
h16 frozen ilm historical	0.40974	61.015	NA	3.387
h8 frozen ilm historical	0.40914	61.104	NA	3.413
h16 frozen i20m historical	0.40995	60.984	NA	3.375
h8 frozen i20m historical	0.40925	61.087	NA	3.448

Run	Coverage	Miss reduction	Raw accuracy	Selected accuracy	Timeliness	P/N_L
h16 frozen ilm	0.583	0.583	0.244	0.244	0.999	1.206
h8 frozen ilm	0.581	0.581	0.244	0.244	1.000	1.201
h8 frozen ilm /64 /MAC0	0.581	0.581	0.244	0.244	1.000	1.201
h8 frozen ilm /64 /MAC16	0.006	0.006	0.008	0.008	0.998	0.412
h8 frozen ilm /64 /MAC4	0.003	0.003	0.006	0.006	0.989	0.227
h16 frozen i20m	0.584	0.584	0.275	0.275	0.999	1.070
h8 frozen i20m	0.577	0.577	0.257	0.257	0.999	1.131
No prefetch	0.000	0.000	NA	NA	NA	0.000
h16 scratch s17	0.507	0.507	0.264	0.264	0.997	0.970
h16 scratch s27	0.471	0.471	0.266	0.266	0.996	0.894
h16 scratch s7	0.541	0.541	0.265	0.266	0.998	1.028
h8 scratch s17	0.574	0.574	0.262	0.262	0.998	1.106
h8 scratch s27	0.578	0.578	0.262	0.264	0.998	1.111
h8 scratch s7	0.571	0.571	0.265	0.265	0.998	1.086
h8 scratch s7 /64 /MAC0	0.571	0.571	0.265	0.265	0.998	1.086
h8 scratch s7 /64 /MAC16	0.006	0.006	0.008	0.008	0.999	0.403
h8 scratch s7 /64 /MAC4	0.002	0.002	0.005	0.005	0.996	0.230
h16 warm ilm s7	0.421	0.421	0.267	0.267	0.996	0.794
h8 warm ilm s7	0.569	0.569	0.268	0.268	0.997	1.072
Stride	0.171	0.171	0.288	0.288	0.995	0.299
h16 frozen ilm	0.583	0.583	0.242	0.242	1.000	1.209
h8 frozen ilm	0.580	0.580	0.306	0.306	1.000	0.952
h16 frozen i20m	0.585	0.585	0.274	0.274	0.999	1.073
h8 frozen i20m	0.576	0.576	0.257	0.257	0.999	1.126

3.1 What the online arms actually observed and processed

Run	Eligible	NN admitted	NN completed	Labels	Positive labels
h16 scratch s17	404,271	404,271	404,271	404,271	62,624
h16 scratch s27	404,271	404,271	404,271	404,271	62,298
h16 scratch s7	404,271	404,271	404,271	404,271	62,731
h8 scratch s17	404,271	404,271	404,271	404,271	62,753
h8 scratch s27	404,271	404,271	404,271	404,271	62,853
h8 scratch s7	404,271	404,271	404,271	404,271	62,706
h16 warm ilm s7	404,271	404,271	404,271	404,271	62,115
h8 warm ilm s7	404,271	404,271	404,271	404,271	62,637

Run	Eligible	Labels	Positive actions	Updates	Exposures	Versions
h16 scratch s17	404,271	404,271	122,922	6,316	404,224	6,316
h16 scratch s27	404,271	404,271	122,261	6,316	404,224	6,316
h16 scratch s7	404,271	404,271	123,177	6,316	404,224	6,316
h8 scratch s17	404,271	404,271	123,226	6,316	404,224	6,316
h8 scratch s27	404,271	404,271	123,418	6,316	404,224	6,316
h8 scratch s7	404,271	404,271	123,127	6,316	404,224	6,316

Run	Eligible	Labels	Positive actions	Updates	Exposures	Versions
h16 warm ilm s7	404,271	404,271	121,852	6,316	404,224	6,316
h8 warm ilm s7	404,271	404,271	122,990	6,316	404,224	6,316

h16 scratch s17: 5,917,450 cumulative cycles saved; final full-window saving -664 cycles. The maximum observed same-lifetime TBPTT span was 41 events, with 0 state evictions, 0 inference input drops and 0 output drops. Available supervision contained 122,922 action atoms, while 404,224 decisions were exposed to completed updates.

The largest observed padded projection batch had 576 positions (actual same-lifetime span 41, versus a configured global batch of 64). Final completed-batch loss was 0.09031 and gradient L2 norm 0.16957. These are optimization diagnostics on received training labels, not useful coverage or a quality reward. Final causal hurdle weights were [0.5916489362716675, 3.2278010845184326].

h16 scratch s27: 5,279,919 cumulative cycles saved; final full-window saving -2,849 cycles. The maximum observed same-lifetime TBPTT span was 41 events, with 0 state evictions, 0 inference input drops and 0 output drops. Available supervision contained 122,261 action atoms, while 404,224 decisions were exposed to completed updates.

The largest observed padded projection batch had 576 positions (actual same-lifetime span 41, versus a configured global batch of 64). Final completed-batch loss was 0.07555 and gradient L2 norm 0.20031. These are optimization diagnostics on received training labels, not useful coverage or a quality reward. Final causal hurdle weights were [0.5910848379135132, 3.244694232940674].

h16 scratch s7: 6,519,115 cumulative cycles saved; final full-window saving -2,317 cycles. The maximum observed same-lifetime TBPTT span was 41 events, with 0 state evictions, 0 inference input drops and 0 output drops. Available supervision contained 123,177 action atoms, while 404,224 decisions were exposed to completed updates.

The largest observed padded projection batch had 576 positions (actual same-lifetime span 41, versus a configured global batch of 64). Final completed-batch loss was 0.08423 and gradient L2 norm 0.20004. These are optimization diagnostics on received training labels, not useful coverage or a quality reward. Final causal hurdle weights were [0.591834306716919, 3.22294807434082].

h8 scratch s17: 6,887,926 cumulative cycles saved; final full-window saving 72,492 cycles. The maximum observed same-lifetime TBPTT span was 41 events, with 0 state evictions, 0 inference input drops and 0 output drops. Available supervision contained 123,226 action atoms, while 404,224 decisions were exposed to completed updates.

The largest observed padded projection batch had 576 positions (actual same-lifetime span 41, versus a configured global batch of 64). Final completed-batch loss was 0.47525 and gradient L2 norm 0.22703. These are optimization diagnostics on received training labels, not useful coverage or a quality reward. Final causal hurdle weights were [0.5918776392936707, 3.22101092338562].

h8 scratch s27: 6,911,684 cumulative cycles saved; final full-window saving 77,795 cycles. The maximum observed same-lifetime TBPTT span was 41 events, with 0 state evictions, 0 inference input drops and 0 output drops. Available supervision contained 123,418 action atoms, while 404,224 decisions were exposed to completed updates.

The largest observed padded projection batch had 576 positions (actual same-lifetime span 41, versus a configured global batch of 64). Final completed-batch loss was 0.47499 and gradient L2 norm 0.20131. These are optimization diagnostics on received training labels, not useful coverage or a quality reward. Final causal hurdle weights were [0.5920509696006775, 3.215885877609253].

h8 scratch s7: 6,860,103 cumulative cycles saved; final full-window saving 66,500 cycles. The maximum observed same-lifetime TBPTT span was 41 events, with 0 state evictions, 0 inference input drops and 0 output drops. Available supervision contained 123,127 action atoms, while 404,224 decisions were exposed to completed updates.

The largest observed padded projection batch had 576 positions (actual same-lifetime span 41, versus a configured global batch of 64). Final completed-batch loss was 0.42241 and gradient L2 norm 0.48178. These are optimization diagnostics on received training labels, not useful coverage or a quality reward. Final causal hurdle weights were [0.5917961597442627, 3.2234253883361816].

h16 warm ilm s7: 4,421,002 cumulative cycles saved; final full-window saving 15,120 cycles. The maximum observed same-lifetime TBPTT span was 41 events, with 0 state evictions, 0 inference input drops and 0 output drops. Available supervision contained 121,852 action atoms, while 404,224 decisions were exposed to completed updates.

The largest observed padded projection batch had 576 positions (actual same-lifetime span 41, versus a configured

global batch of 64). Final completed-batch loss was 0.05978 and gradient L2 norm 0.14508. These are optimization diagnostics on received training labels, not useful coverage or a quality reward. Final causal hurdle weights were [0.5907686352729797, 3.2542548179626465].

h8 warm ilm s7: 6,895,502 cumulative cycles saved; final full-window saving 69,627 cycles. The maximum observed same-lifetime TBPTT span was 41 events, with 0 state evictions, 0 inference input drops and 0 output drops. Available supervision contained 122,990 action atoms, while 404,224 decisions were exposed to completed updates.

The largest observed padded projection batch had 576 positions (actual same-lifetime span 41, versus a configured global batch of 64). Final completed-batch loss was 0.37132 and gradient L2 norm 0.32214. These are optimization diagnostics on received training labels, not useful coverage or a quality reward. Final causal hurdle weights were [0.5916748642921448, 3.2270281314849854].

3.2 Preserved metric definitions

With M_0 matched no-prefetch L2 LOAD misses, M method misses, N_L L2 LOADs, P requests, I_{pf} issued requests, Q PQ merges, U useful prefetches, and L late prefetches:

$$\text{coverage} = U/M_0, \quad \text{miss reduction} = (M_0 - M)/M_0, \quad \text{raw accuracy} = U/I_{pf},$$

$$\text{selected accuracy} = U/(I_{pf} - Q), \quad \text{legacy timeliness} = U/(U + L), \quad \text{pressure} = P/N_L.$$

The original cache’s useful-bit accounting can include a first RFO demand, while N_L and M retain their LOAD scope. Original printed LOAD misses increment at real fills; accepted lookup/callback miss counts are additional fields, not substitutes for M . Merged demands can therefore make callback and printed-fill counts differ. These legacy definitions are preserved. Neither $1 - \text{accuracy}$ nor unused eviction is called direct cache pollution. Counter deltas in windows measure activity within those windows; an issue and its later usefulness may cross a window boundary, so a window accuracy can be undefined even when a useful event occurred.

The observed h8 scratch startup samples were at 0.1M: -3,568 saved cycles, 1,659 callbacks, and 25 completed updates; at 0.5M: 107,914–135,701 saved cycles, 8,100 callbacks, and 126 completed updates; at 1M: 257,123–284,360 saved cycles, 16,196 callbacks, and 253 completed updates. These sampled values do not identify an exact event-level crossing time.

The observed h16 scratch startup samples were at 0.1M: 3,632–10,047 saved cycles, 1,659 callbacks, and 25 completed updates; at 0.5M: 130,587–136,696 saved cycles, 8,100 callbacks, and 126 completed updates; at 1M: 278,070–285,305 saved cycles, 16,196 callbacks, and 253 completed updates. These sampled values do not identify an exact event-level crossing time.

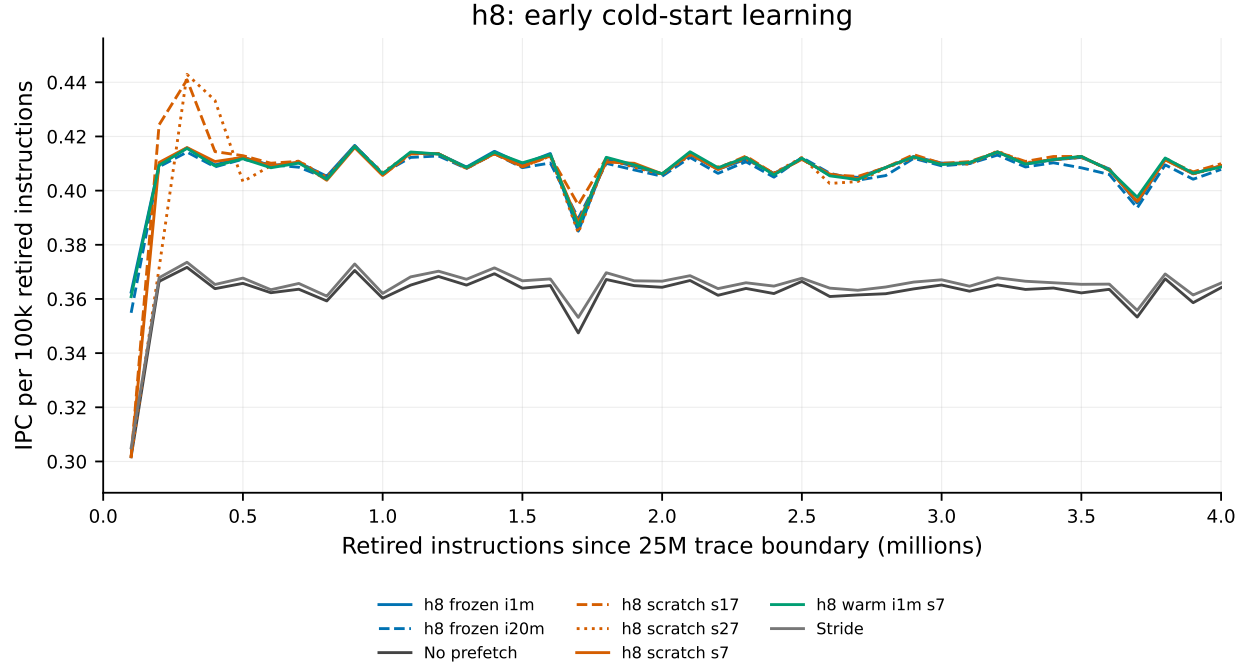


Figure 1: Early measurements use 100k-instruction intervals. They describe the startup trajectory; the predeclared learning milestones continue to use three consecutive 1M windows.

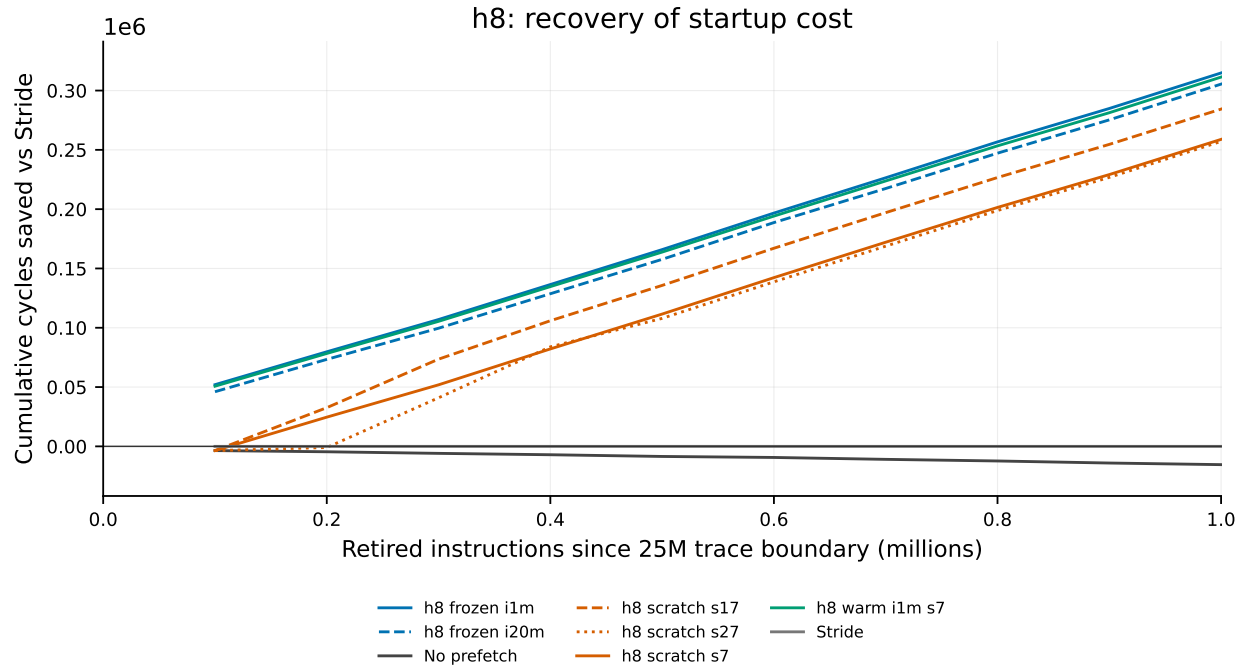


Figure 2: The first million instructions retains the initially negative region. A single positive sampled endpoint is descriptive and does not replace the three-window stability rule.

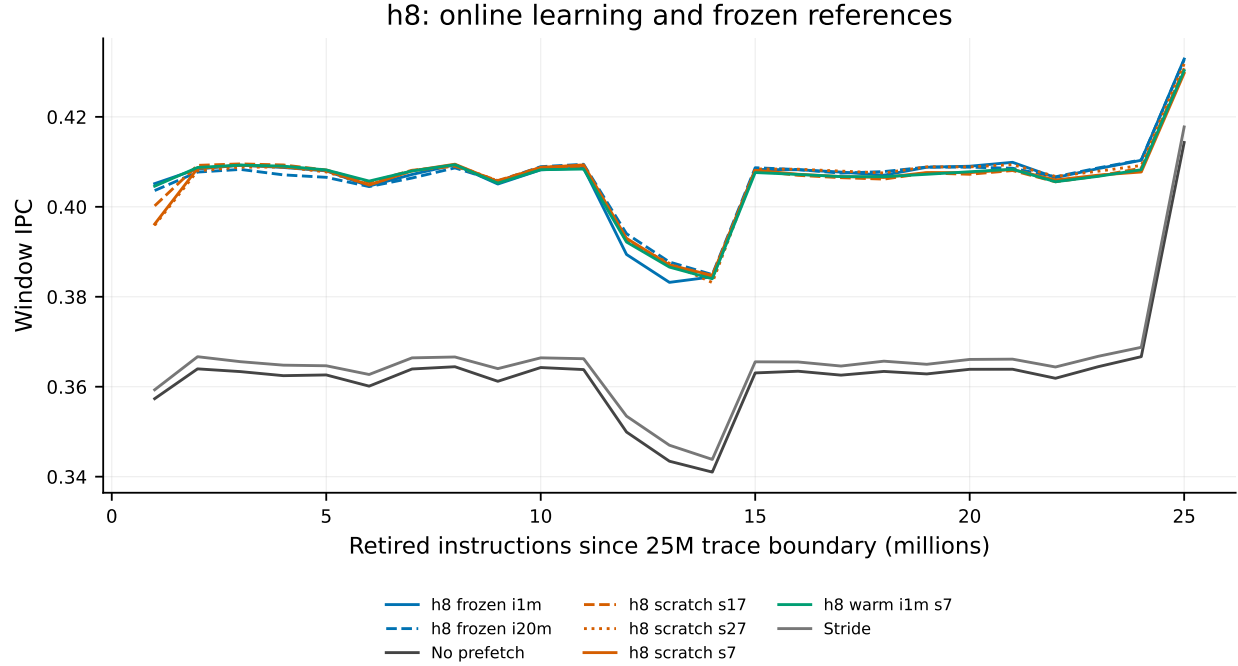


Figure 3: Cold-start functional measurements. Each point is a 1M-retired-instruction window. Scratch seeds 7, 17, and 27 are separate runs; pretrained checkpoints have only seed 7. Missing or undefined ordinates are not interpolated.

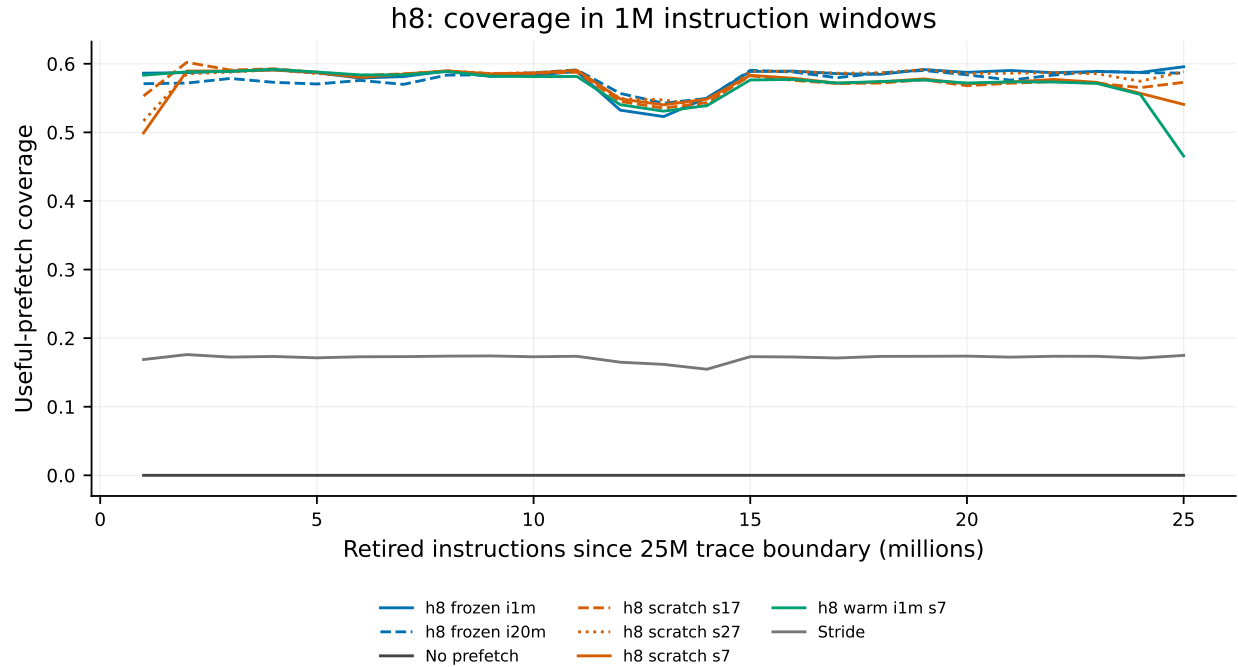


Figure 4: Cold-start functional measurements. Each point is a 1M-retired-instruction window. Scratch seeds 7, 17, and 27 are separate runs; pretrained checkpoints have only seed 7. Missing or undefined ordinates are not interpolated.

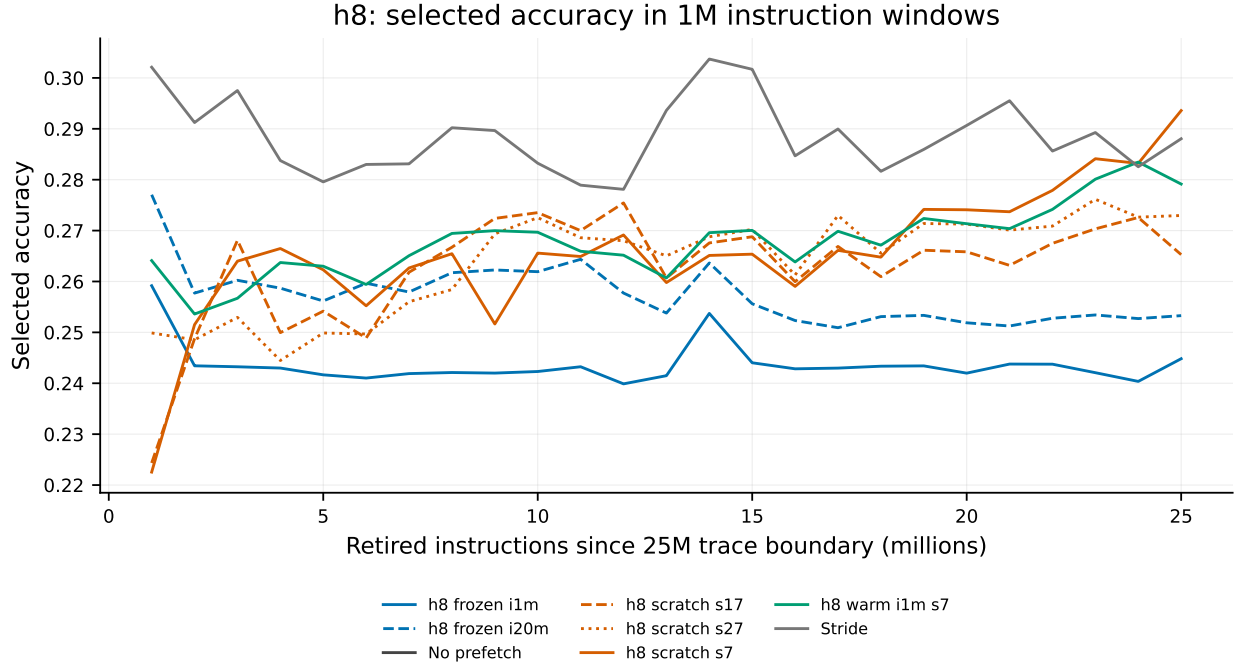


Figure 5: Cold-start functional measurements. Each point is a 1M-retired-instruction window. Scratch seeds 7, 17, and 27 are separate runs; pretrained checkpoints have only seed 7. Missing or undefined ordinates are not interpolated.

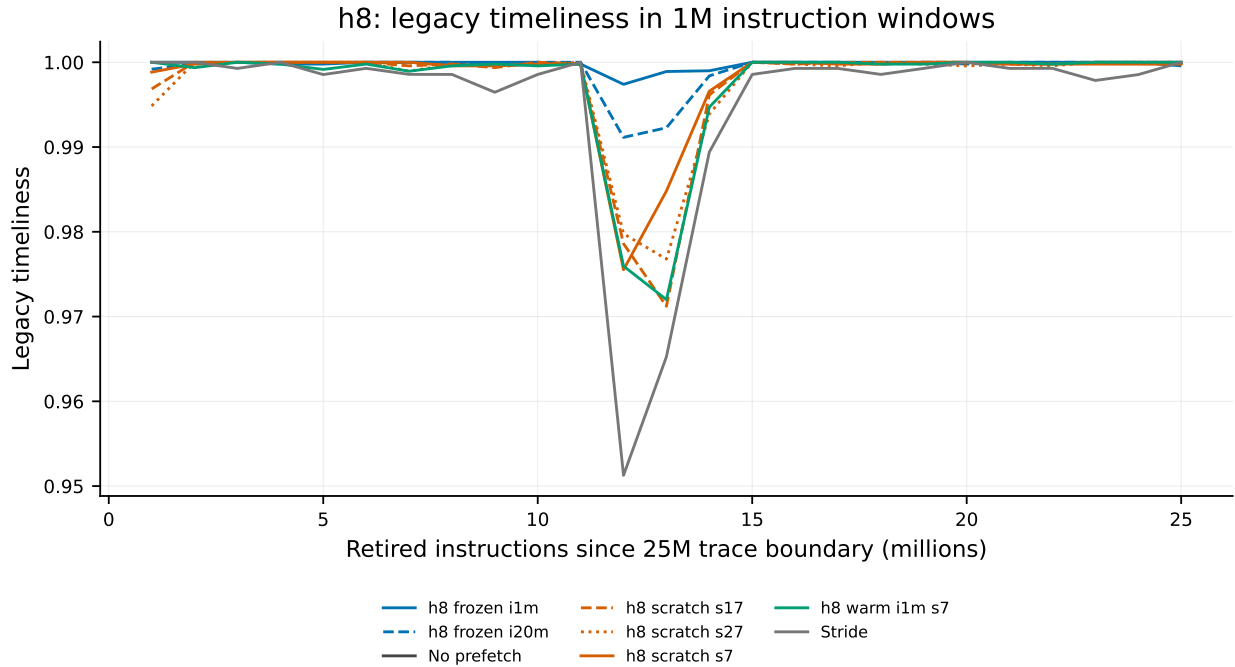


Figure 6: Cold-start functional measurements. Each point is a 1M-retired-instruction window. Scratch seeds 7, 17, and 27 are separate runs; pretrained checkpoints have only seed 7. Missing or undefined ordinates are not interpolated.

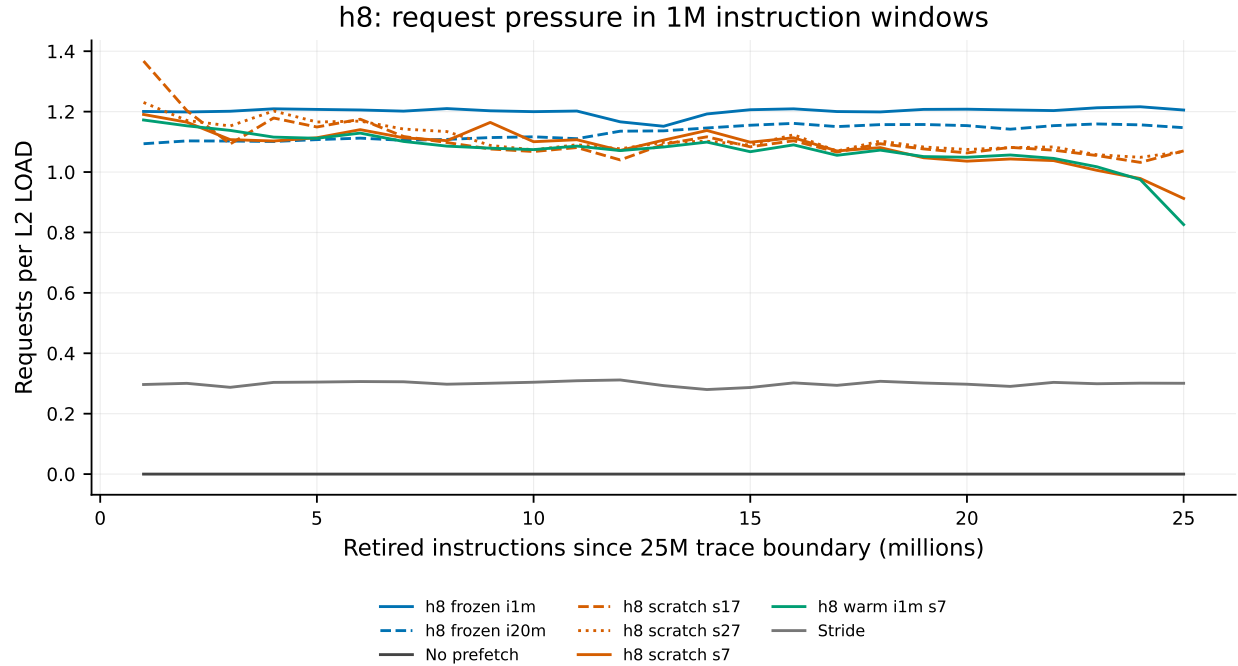


Figure 7: Cold-start functional measurements. Each point is a 1M-retired-instruction window. Scratch seeds 7, 17, and 27 are separate runs; pretrained checkpoints have only seed 7. Missing or undefined ordinates are not interpolated.

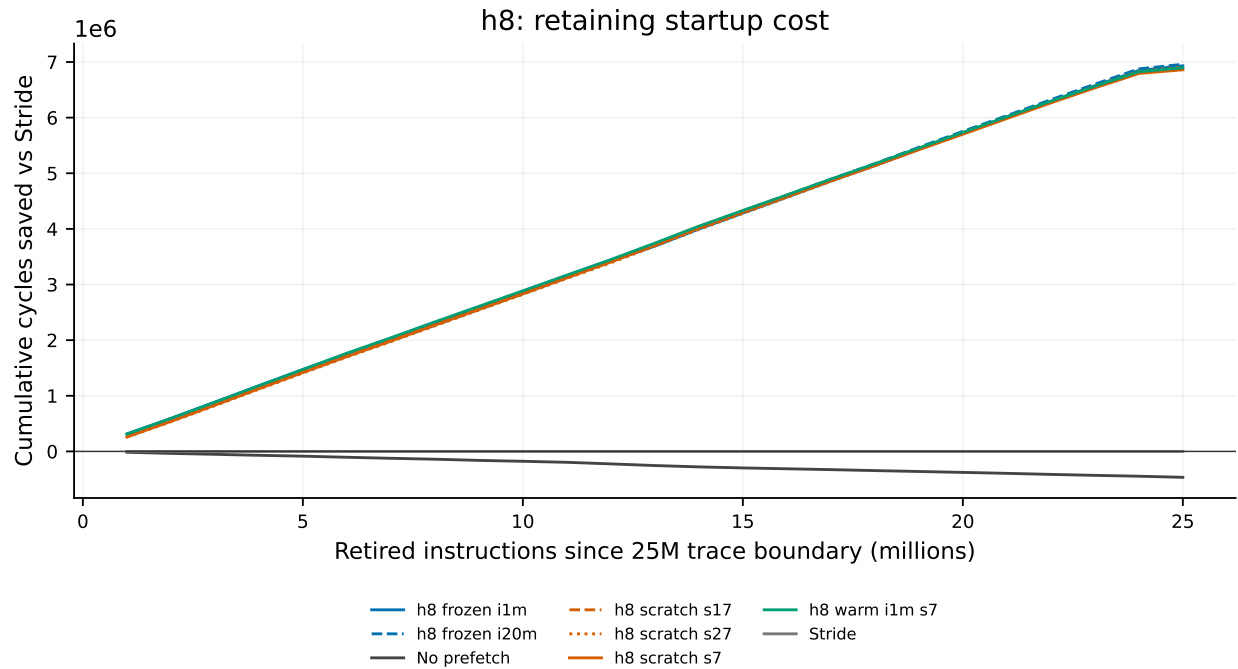


Figure 8: Positive cumulative saving means the run has recovered its startup cost relative to the matched cold-start Stride run. A good late window alone does not establish an end-to-end improvement.

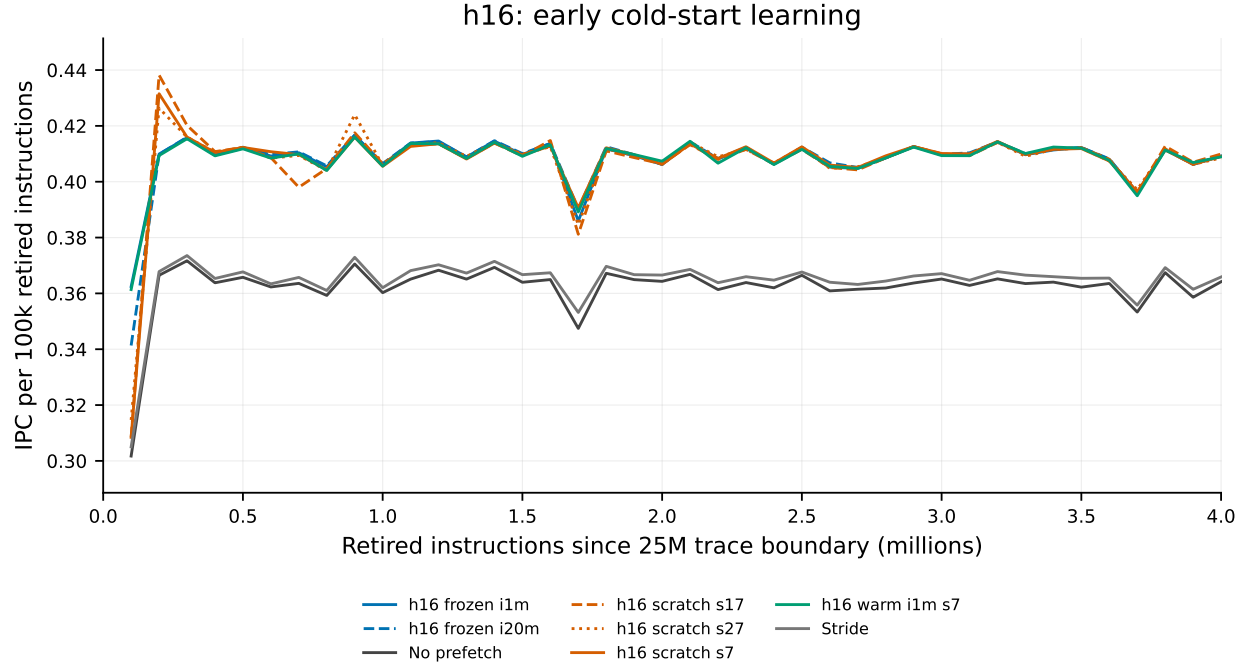


Figure 9: Early measurements use 100k-instruction intervals. They describe the startup trajectory; the predeclared learning milestones continue to use three consecutive 1M windows.

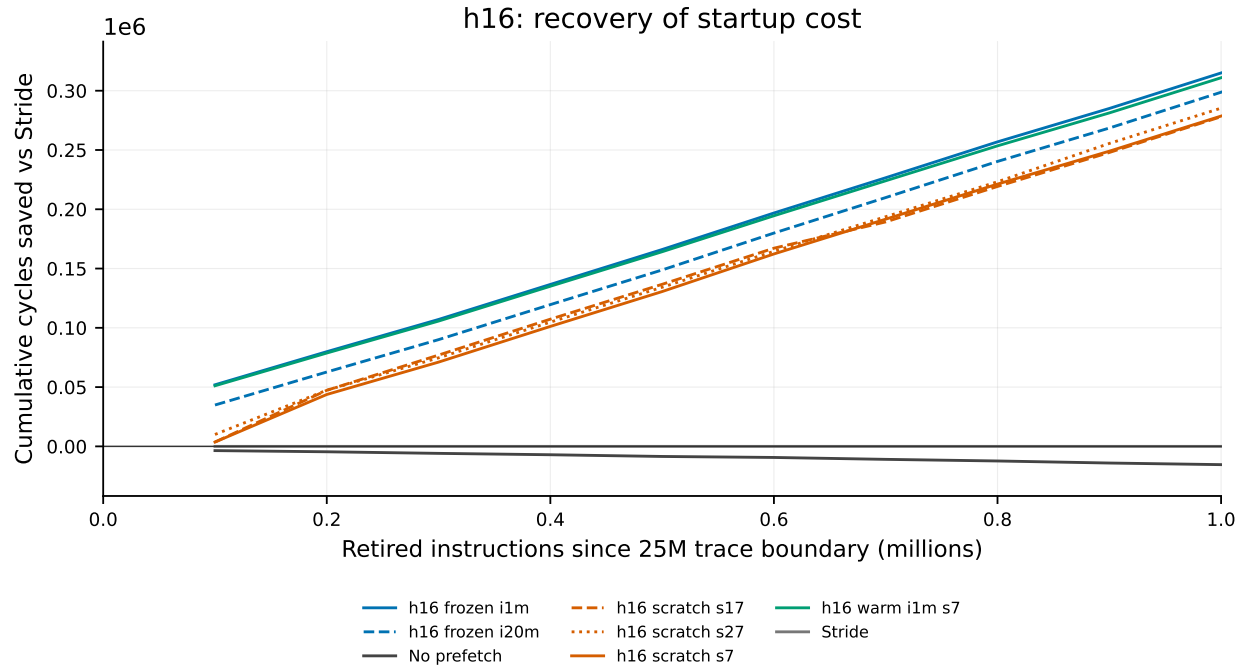


Figure 10: The first million instructions retains the initially negative region. A single positive sampled endpoint is descriptive and does not replace the three-window stability rule.

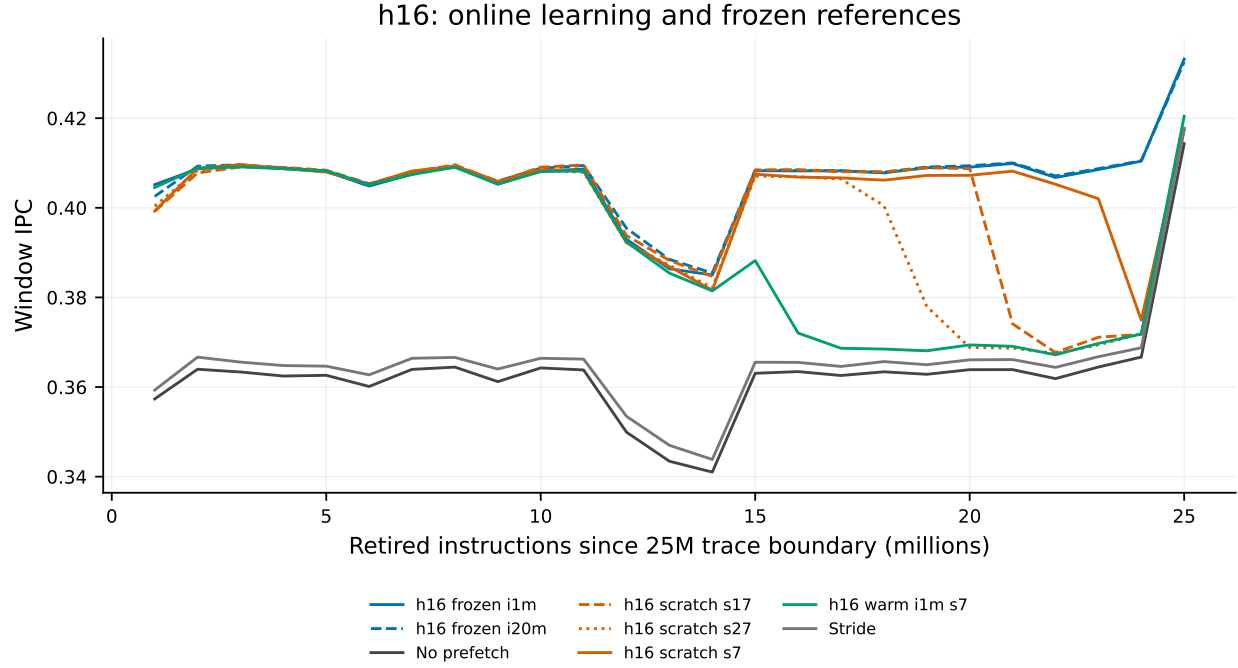


Figure 11: Cold-start functional measurements. Each point is a 1M-retired-instruction window. Scratch seeds 7, 17, and 27 are separate runs; pretrained checkpoints have only seed 7. Missing or undefined ordinates are not interpolated.

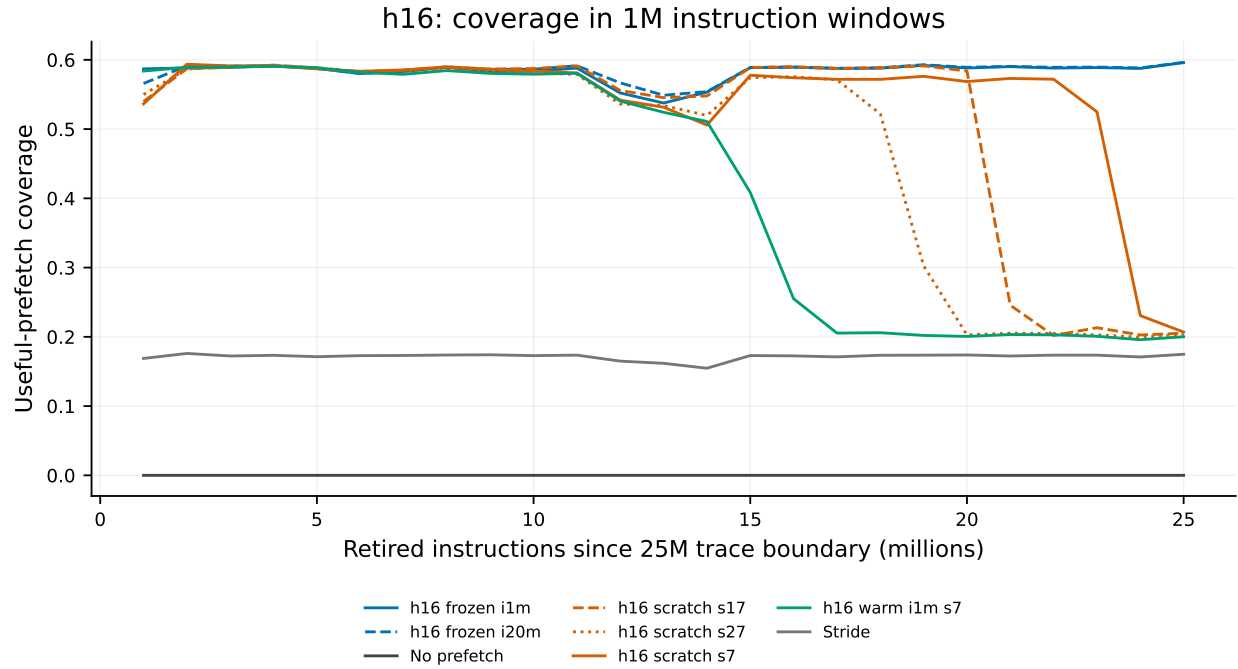


Figure 12: Cold-start functional measurements. Each point is a 1M-retired-instruction window. Scratch seeds 7, 17, and 27 are separate runs; pretrained checkpoints have only seed 7. Missing or undefined ordinates are not interpolated.

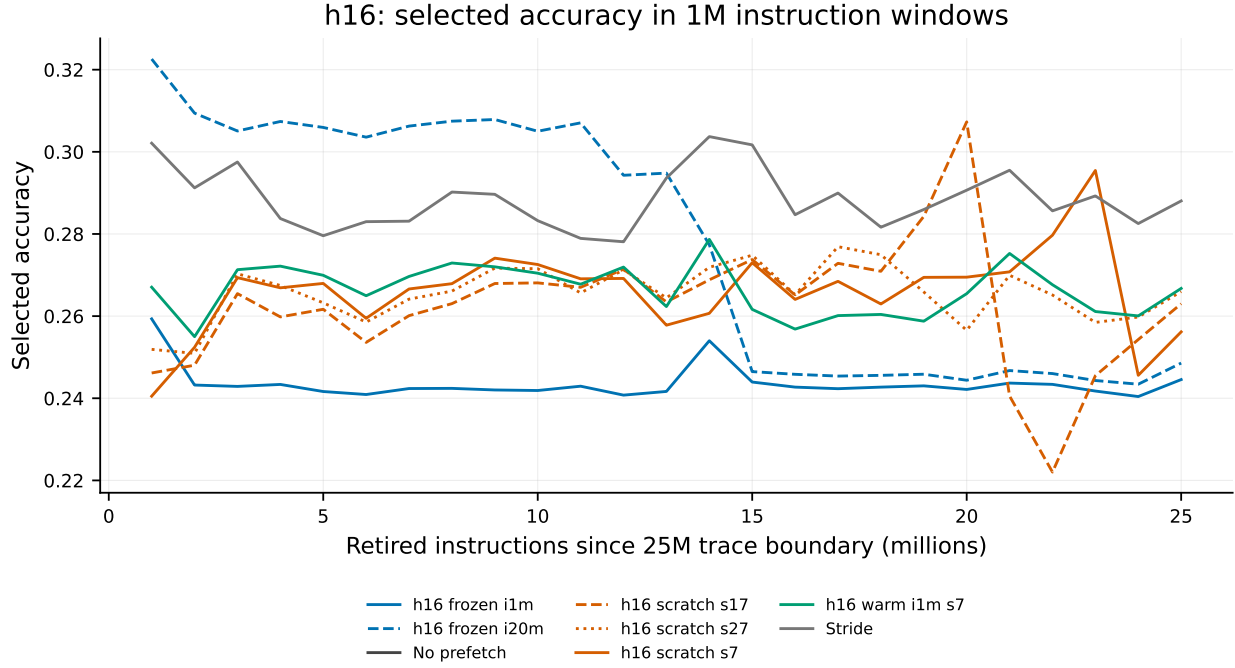


Figure 13: Cold-start functional measurements. Each point is a 1M-retired-instruction window. Scratch seeds 7, 17, and 27 are separate runs; pretrained checkpoints have only seed 7. Missing or undefined ordinates are not interpolated.

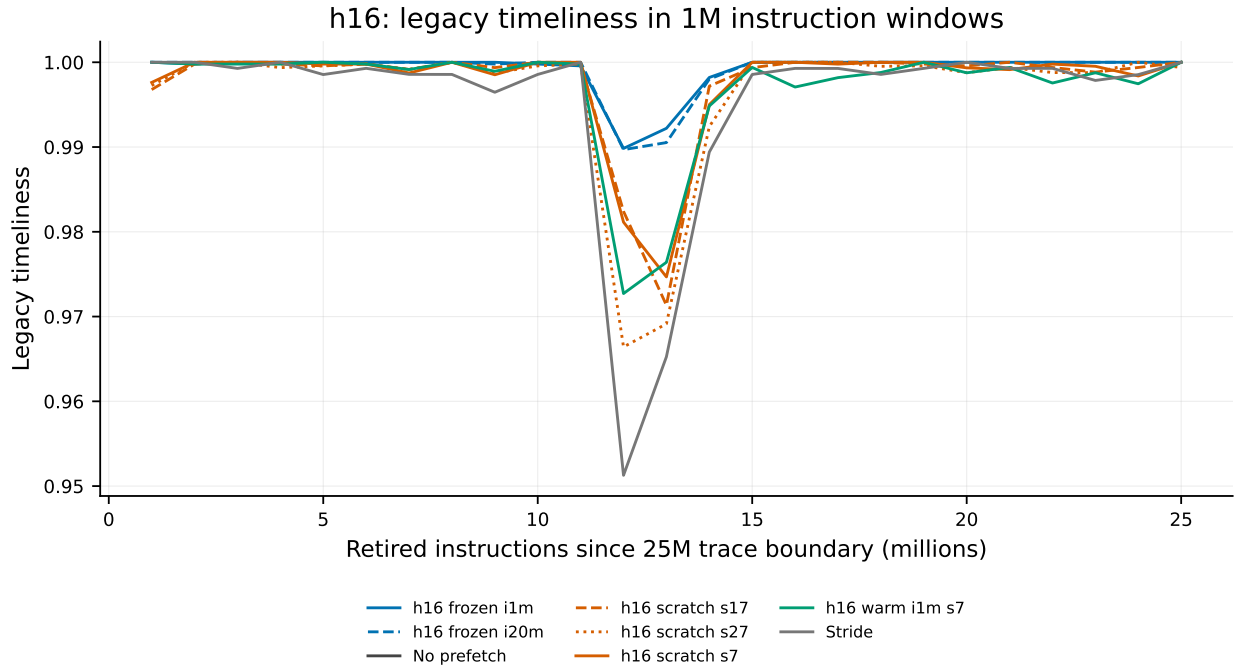


Figure 14: Cold-start functional measurements. Each point is a 1M-retired-instruction window. Scratch seeds 7, 17, and 27 are separate runs; pretrained checkpoints have only seed 7. Missing or undefined ordinates are not interpolated.

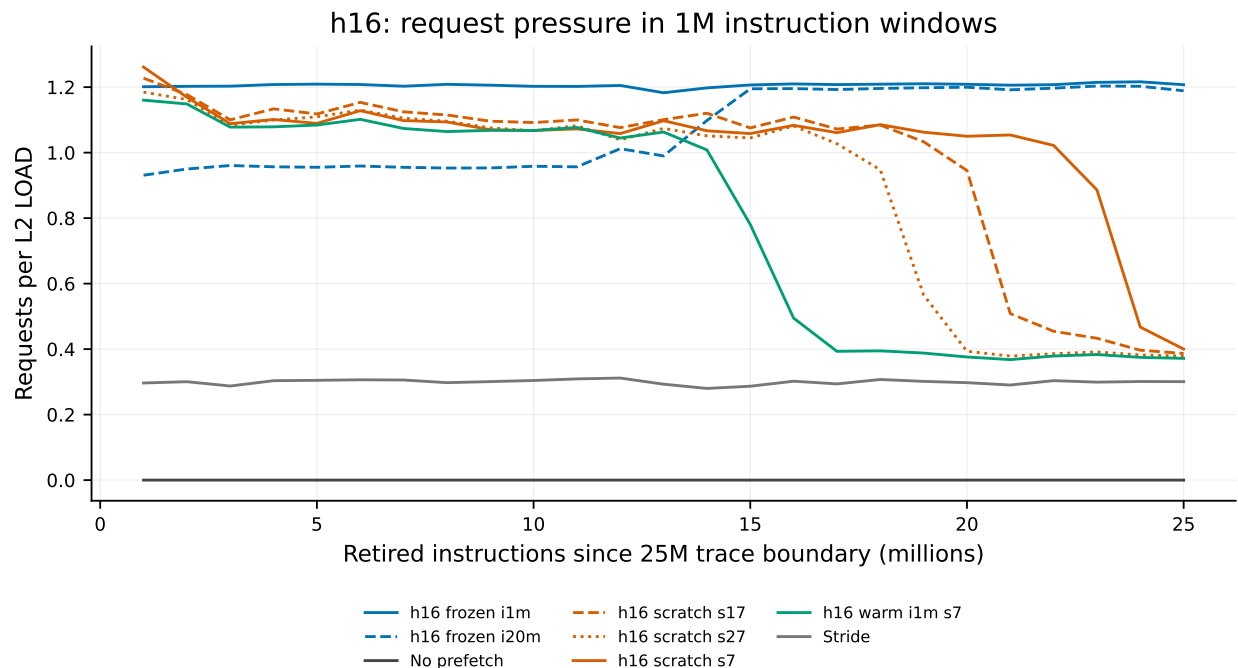


Figure 15: Cold-start functional measurements. Each point is a 1M-retired-instruction window. Scratch seeds 7, 17, and 27 are separate runs; pretrained checkpoints have only seed 7. Missing or undefined ordinates are not interpolated.

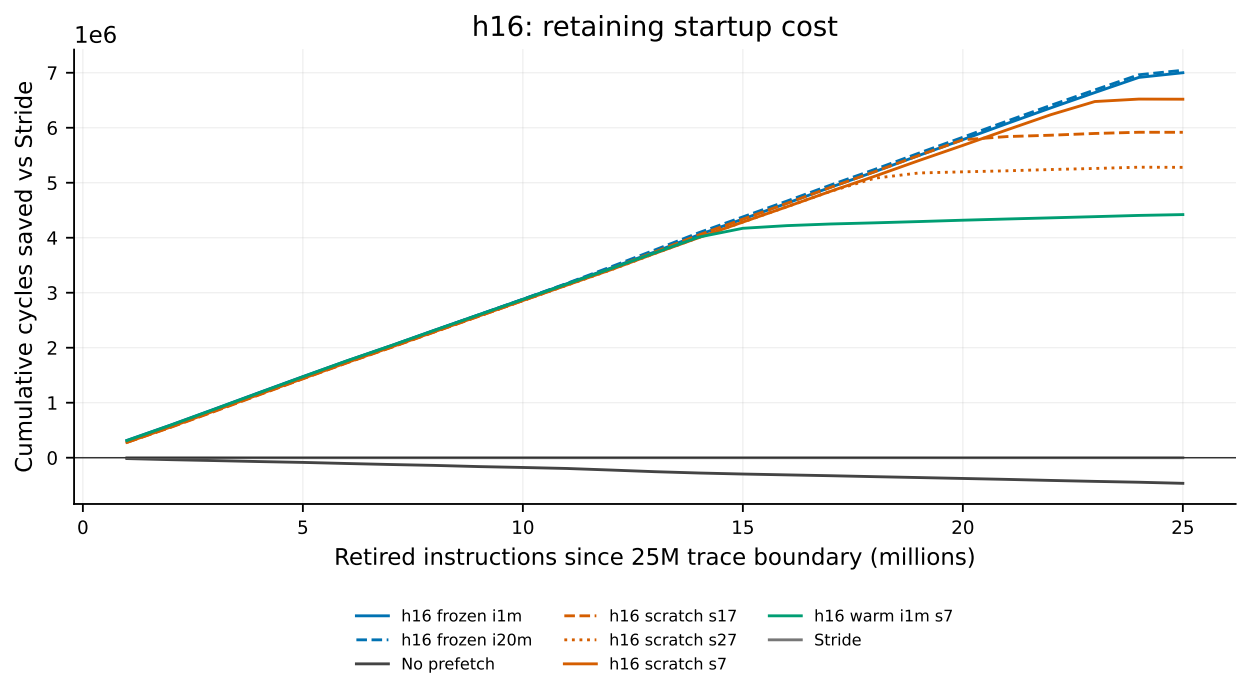


Figure 16: Positive cumulative saving means the run has recovered its startup cost relative to the matched cold-start Stride run. A good late window alone does not establish an end-to-end improvement.

4 Observed learning milestones

The descriptive stability rule is three consecutive 1M-retired-instruction endpoints with positive cumulative cycle savings versus Stride. The separate frozen target requires window IPC at least 99% of same-H frozen-ilm IPC in three consecutive 1M windows. Tables give the first qualifying window endpoint and the confirmation endpoint. This rule is not an online controller or a statistical equivalence test. Positive window streaks are retained separately in CSV; they do not imply startup has been recovered.

Run	Milestone	First M	Confirm M	Updates first/conf.	Callbacks first/conf.
h16 scratch s17	Saved cycles	1.0	3.0	253/757	16,196/48,453
h16 scratch s17	99% frozen	2.0	4.0	505/1,008	32,320/64,553
h16 scratch s27	Saved cycles	1.0	3.0	253/757	16,196/48,453
h16 scratch s27	99% frozen	2.0	4.0	505/1,008	32,320/64,553
h16 scratch s7	Saved cycles	1.0	3.0	253/757	16,196/48,453
h16 scratch s7	99% frozen	2.0	4.0	505/1,008	32,320/64,553
h8 scratch s17	Saved cycles	1.0	3.0	253/757	16,196/48,453
h8 scratch s17	99% frozen	2.0	4.0	505/1,008	32,320/64,553
h8 scratch s27	Saved cycles	1.0	3.0	253/757	16,196/48,453
h8 scratch s27	99% frozen	2.0	4.0	505/1,008	32,320/64,553
h8 scratch s7	Saved cycles	1.0	3.0	253/757	16,196/48,453
h8 scratch s7	99% frozen	2.0	4.0	505/1,008	32,320/64,553
h16 warm ilm s7	Saved cycles	1.0	3.0	253/757	16,196/48,453
h16 warm ilm s7	99% frozen	1.0	3.0	253/757	16,196/48,453
h8 warm ilm s7	Saved cycles	1.0	3.0	253/757	16,196/48,453
h8 warm ilm s7	99% frozen	1.0	3.0	253/757	16,196/48,453

Run	Milestone	Coverage	Selected accuracy	Timeliness	P/N_L
h16 scratch s17	Saved cycles	0.5398	0.2462	0.9968	1.2275
h16 scratch s17	99% frozen	0.5875	0.2481	0.9998	1.1773
h16 scratch s27	Saved cycles	0.5500	0.2519	0.9973	1.1847
h16 scratch s27	99% frozen	0.5866	0.2509	1.0000	1.1620
h16 scratch s7	Saved cycles	0.5364	0.2405	0.9976	1.2614
h16 scratch s7	99% frozen	0.5937	0.2524	1.0000	1.1692
h8 scratch s17	Saved cycles	0.5531	0.2243	0.9969	1.3675
h8 scratch s17	99% frozen	0.6022	0.2488	1.0000	1.2033
h8 scratch s27	Saved cycles	0.5167	0.2499	0.9948	1.2304
h8 scratch s27	99% frozen	0.5853	0.2485	1.0000	1.1707
h8 scratch s7	Saved cycles	0.4991	0.2225	0.9988	1.1906
h8 scratch s7	99% frozen	0.5894	0.2515	0.9998	1.1650
h16 warm ilm s7	Saved cycles	0.5838	0.2670	1.0000	1.1603
h16 warm ilm s7	99% frozen	0.5838	0.2670	1.0000	1.1603
h8 warm ilm s7	Saved cycles	0.5834	0.2641	1.0000	1.1723
h8 warm ilm s7	99% frozen	0.5834	0.2641	1.0000	1.1723

The corresponding coverage, selected accuracy, legacy timeliness, request pressure, supervised decisions, exposures, and published versions are in `learning_milestones.csv`; they remain separate quantities rather than a composite score.

5 Cache occupancy, turnover, and lifecycle outcomes

Occupancy counts simultaneously valid L2 lines. A valid-to-valid replacement changes the resident line without increasing fullness. Demand LOAD, RFO, prefetch, and writeback fills are separate, alongside replacement count and useful-line lifetimes. Fullness is not proof of a warmed working set or a deadline after which prefetching is useless.

The active L2 geometry is 512 sets $\times$ 8 ways $\times$ 64 bytes per line: 4,096 lines, 262,144 bytes (256 KiB).

Run	Occupancy	Instructions	Cycles
h8 frozen ilm	50%	176,919	464,590
h8 frozen ilm	90%	388,585	977,223
h8 frozen ilm	95%	417,448	1,046,864

Run	Occupancy	Instructions	Cycles
h8 frozen ilm	99%	460,465	1,152,325
h8 frozen ilm	100%	536,006	1,334,484
No prefetch	50%	180,391	551,809
No prefetch	90%	392,060	1,126,716
No prefetch	95%	420,440	1,203,868
No prefetch	99%	461,565	1,317,623
No prefetch	100%	536,244	1,520,774
h8 scratch s7	50%	178,091	523,505
h8 scratch s7	90%	389,162	1,032,946
h8 scratch s7	95%	418,460	1,103,497
h8 scratch s7	99%	460,685	1,207,191
h8 scratch s7	100%	536,006	1,389,058
h8 warm ilm s7	50%	176,553	465,093
h8 warm ilm s7	90%	389,020	980,026
h8 warm ilm s7	95%	417,660	1,049,035
h8 warm ilm s7	99%	460,625	1,154,770
h8 warm ilm s7	100%	536,006	1,336,799
Stride	50%	180,091	546,618
Stride	90%	391,562	1,118,246
Stride	95%	420,040	1,195,372
Stride	99%	461,565	1,309,716
Stride	100%	536,244	1,512,126

The supplementary `issue_cohorts.csv` tracks actual unique PQ enqueues by issue-instruction cohort. Terminal outcomes are timely first demand, late first demand, unused eviction, redundant cache hit, or redundant in-flight request. Pending and resident-unused requests remain censored. These exclusive cohort outcomes are separate from legacy issued/merged counters and are not relabeled as the legacy accuracy or timeliness metrics.

Run	LOAD fills	RFO fills	PF fills	WB fills	Replacements	Useful lifetimes
h8 frozen ilm	85,475	2	121,310	0	202,691	199,717
No prefetch	203,764	2	0	0	199,670	199,670
h8 scratch s7	87,419	2	120,129	0	203,454	199,717
h8 warm ilm s7	87,778	2	119,501	0	203,185	199,716
Stride	168,865	2	35,439	0	200,210	199,675

Run	Mean closed line lifetime	Mean useful closed lifetime	Mean PF fill to first use
h8 frozen ilm	1,142,718.8		1,143,143.0
No prefetch	1,303,645.7		1,303,645.7
h8 scratch s7	1,141,630.2		1,141,235.8
h8 warm ilm s7	1,143,321.8		1,143,359.6
Stride	1,291,315.4		1,291,631.5

Lifetime means are simulated cycles, not fill counts. Closed resident lifetimes end on eviction or invalidation; the useful subset served a demand or RFO at fill or during residence. Prefetch fill-to-first-use measures the first demand on a prefetched line. Still-resident line lifetimes are censored rather than assigned an end time; historical lines whose birth predates measurement have no fabricated start time.

Run	Learning endpoint	M instr.	Occupancy	Replacements	Turnovers	Updates
h16 scratch s17	Saving first	1.0	100.0%	4,581	1.12	253
h16 scratch s17	Saving confirm	3.0	100.0%	20,731	5.06	757
h16 scratch s17	99% frozen first	2.0	100.0%	12,649	3.09	505

Run	Learning endpoint	M instr.	Occupancy	Replacements	Turnovers	Updates
h16 scratch s17	99% frozen confirm	4.0	100.0%	28,835	7.04	1,008
h16 scratch s27	Saving first	1.0	100.0%	4,576	1.12	253
h16 scratch s27	Saving confirm	3.0	100.0%	20,754	5.07	757
h16 scratch s27	99% frozen first	2.0	100.0%	12,654	3.09	505
h16 scratch s27	99% frozen confirm	4.0	100.0%	28,872	7.05	1,008
h16 scratch s7	Saving first	1.0	100.0%	4,576	1.12	253
h16 scratch s7	Saving confirm	3.0	100.0%	20,796	5.08	757
h16 scratch s7	99% frozen first	2.0	100.0%	12,671	3.09	505
h16 scratch s7	99% frozen confirm	4.0	100.0%	28,926	7.06	1,008
h8 scratch s17	Saving first	1.0	100.0%	4,580	1.12	253
h8 scratch s17	Saving confirm	3.0	100.0%	20,814	5.08	757
h8 scratch s17	99% frozen first	2.0	100.0%	12,670	3.09	505
h8 scratch s17	99% frozen confirm	4.0	100.0%	28,936	7.06	1,008
h8 scratch s27	Saving first	1.0	100.0%	4,553	1.11	253
h8 scratch s27	Saving confirm	3.0	100.0%	20,670	5.05	757
h8 scratch s27	99% frozen first	2.0	100.0%	12,605	3.08	505
h8 scratch s27	99% frozen confirm	4.0	100.0%	28,766	7.02	1,008
h8 scratch s7	Saving first	1.0	100.0%	4,560	1.11	253
h8 scratch s7	Saving confirm	3.0	100.0%	20,795	5.08	757
h8 scratch s7	99% frozen first	2.0	100.0%	12,665	3.09	505
h8 scratch s7	99% frozen confirm	4.0	100.0%	28,915	7.06	1,008
h16 warm ilm s7	Saving first	1.0	100.0%	4,558	1.11	253
h16 warm ilm s7	Saving confirm	3.0	100.0%	20,797	5.08	757
h16 warm ilm s7	99% frozen first	1.0	100.0%	4,558	1.11	253
h16 warm ilm s7	99% frozen confirm	3.0	100.0%	20,797	5.08	757
h8 warm ilm s7	Saving first	1.0	100.0%	4,562	1.11	253
h8 warm ilm s7	Saving confirm	3.0	100.0%	20,830	5.09	757
h8 warm ilm s7	99% frozen first	1.0	100.0%	4,562	1.11	253
h8 warm ilm s7	99% frozen confirm	3.0	100.0%	20,830	5.09	757

The table places cache turnover beside each first and confirmed learning endpoint. A full cache continues replacing lines while learning and useful prefetching continue. Occupancy is neither a working-set warmth test nor a learning deadline.

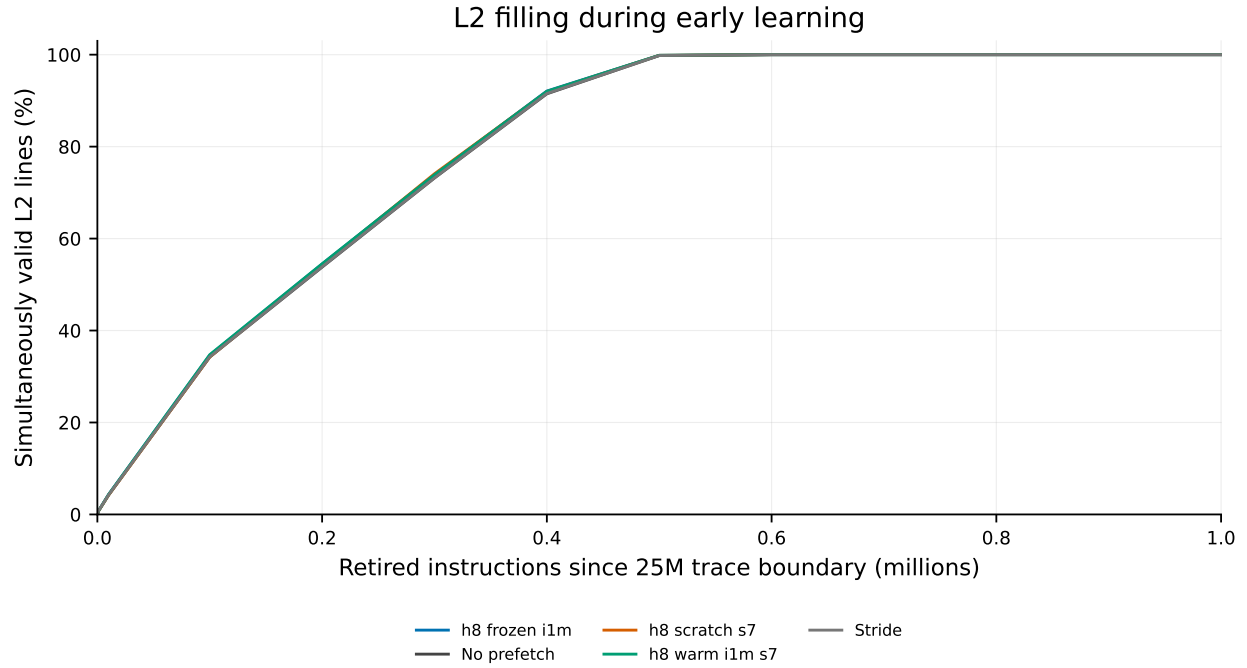


Figure 17: The trace-slice origin is 25M target instructions; this plot expands its first million instructions. Exact fill-event first-attainment records, rather than cumulative fill counts, determine the occupancy milestones.

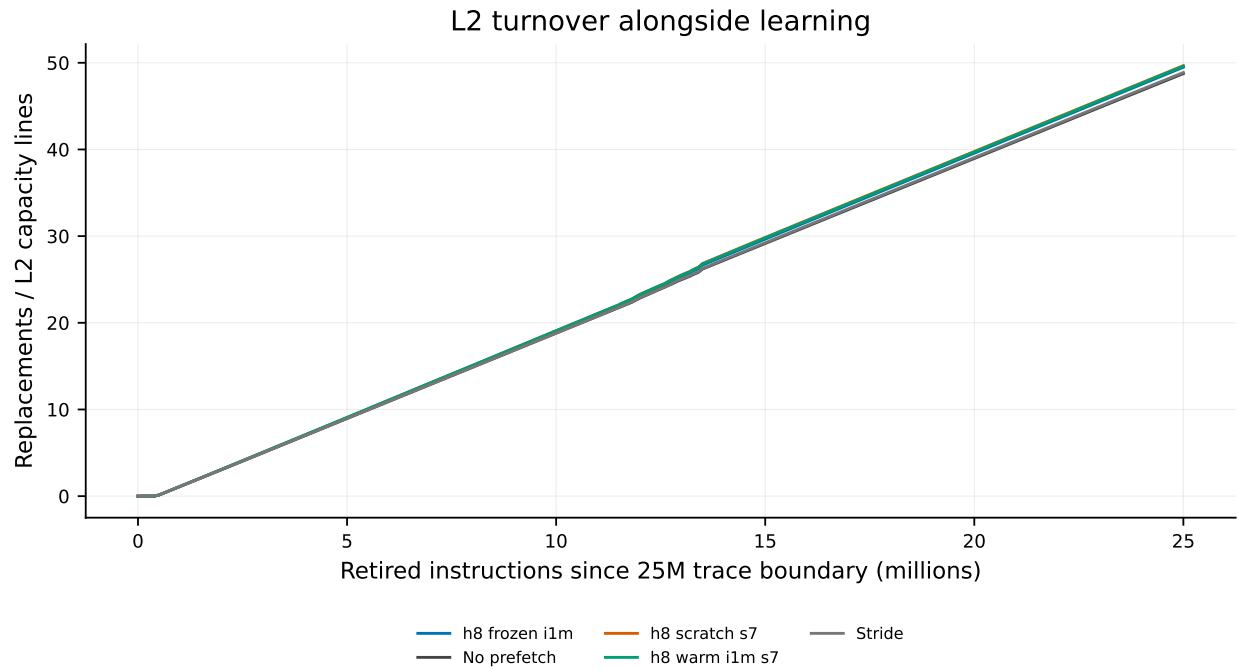


Figure 18: Turnover is cumulative valid-to-valid replacements normalized by actual L2 line capacity. It is distinct from simultaneous occupancy and from useful-prefetch coverage.

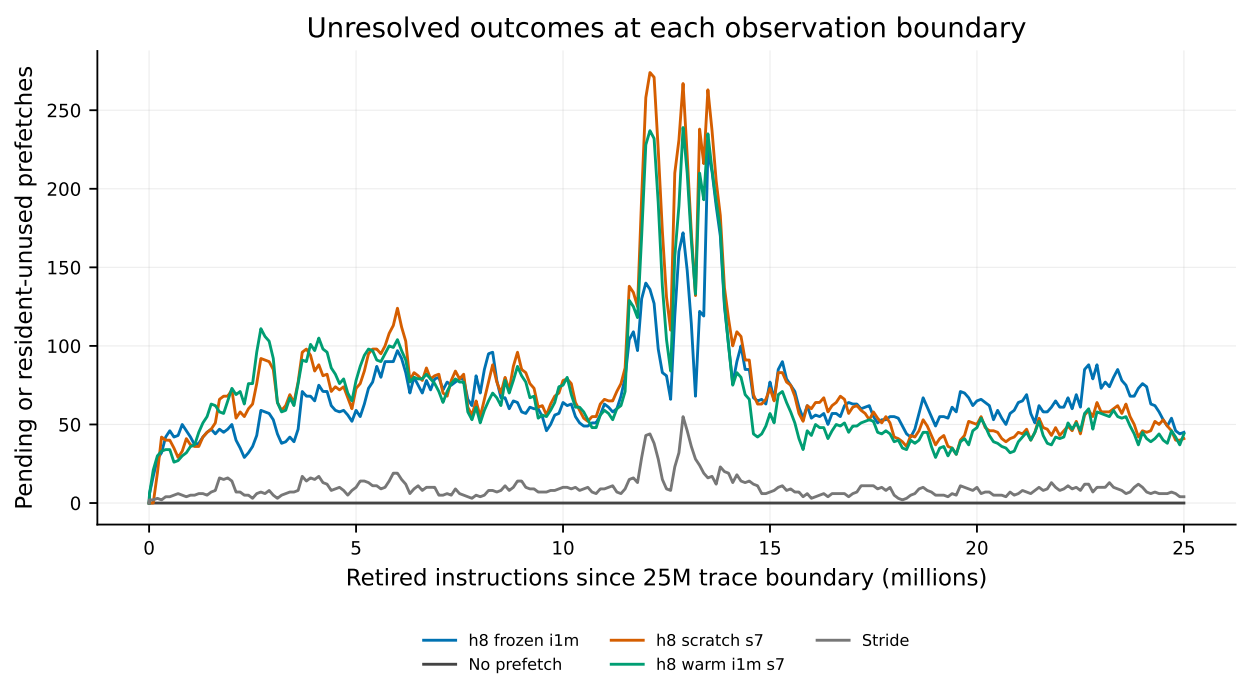


Figure 19: These outcomes remain censored at the displayed boundary. Their later outcomes do not retroactively redefine a window’s observed state; final issue-cohort data are provided separately.

6 Storage and timing sensitivity

Unbounded state and zero modeled latency are algorithmic controls. Finite arms use the same exact-PC table and queue limits for frozen and online inference. Equal entry counts do not imply equal byte cost. FP32 weights are 7,632 bytes (7.453 KiB) for h8 and 20,880 bytes (20.391 KiB) for h16. A bare h/c pair requires $8H$ bytes per entry before tags and replacement metadata; a 64-entry table therefore uses 4,096 or 8,192 h/c bytes. No quantized-model measurement is claimed.

The finite-resource sensitivity cases are architectural assumptions, not measured silicon latency, area, power, frequency, or PPA. Inference work depends on the actual model and decoded K . Training, active/training weight copies, optimizer moments, publication/copy work, queues, and the shadow teacher must be counted. Host CPU time is not converted into simulated stall cycles. Simulator/CPU co-simulation is not completed on-chip training hardware.

6.1 Predeclared architectural cost assumptions

There are separate serial inference and training engines. The two finite cases provide $b \in \{4, 16\}$ MACs/cycle with matched weight bandwidth $4b \in \{16, 64\}$ bytes/cycle. Each engine has one nonlinear unit at four cycles per operation. Dense, nonlinear, and control phases serialize conservatively. The matched weight-read bandwidth supplies one FP32 weight operand per MAC concurrently within the dense phase; it is not a second serialized full-weight-read charge. The additional bandwidth term below counts scratch/state traffic. The effective inference initiation interval equals its service occupancy because there is one inference in flight; the input queue has 16 entries, output queue 32 addresses, and output issue bandwidth one request/cycle. The learned positive count is not clamped to Stride’s degree: the separate 32-address decoder resource bound drops excess output explicitly.

Let E denote the learned positive hurdle and K the actually decoded count after explicit resource limits. Inference work is

$$F_{\text{infer}} = 128H + 8H^2 + 2H + EH + K(3H^2 + 4H), \quad Z_{\text{infer}} = 6H + E + K(3H + 1).$$

Its service time is $\lceil F_{\text{infer}}/b \rceil + 4Z_{\text{infer}} + 8 + \lceil S/4 \rceil + 4K + \lceil T_{\text{infer}}/(4b) \rceil$, where S is live state entries and $T_{\text{infer}} = 4(128 + 4H) + 24K$ bytes accounts for scratch/state/output traffic. This is a sensitivity model, not a measured pipeline.

For each 64-decision batch, R is actual padded projection positions, n_+ the number of positive decisions, and A the teacher action count. $P_\theta = 11H^2 + 150H + 4$ is parameter count. The actual forward MAC estimate and conservative backward/optimizer costs are

$$F_{\text{train}} = 128HR + 64(8H^2 + 2H) + n_+H + A(3H^2 + 4H),$$

$$Z_{\text{train}} = RH + 320H + 3AH + 192 + n_+ + 2P_\theta.$$

Training costs $\lceil 3F_{\text{train}}/b \rceil + 12Z_{\text{train}} + 512 + 12P_\theta + \lceil T_{\text{train}}/(4b) \rceil$ cycles, with $T_{\text{train}} = 32P_\theta + 4(128R + 128H + AH)$ bytes. The backward MAC bound is twice forward, so total modeled forward/backward work is three times forward; it is not a profiler measurement of PyTorch kernels. Adam/state movement and parameter copying are counted separately. Publication costs $\lceil 4P_\theta/(4b) \rceil + 8$ cycles. One update may be in flight and 64 additional decisions pending; a full pending queue drops training admission explicitly.

The shadow teacher assumes a separate 24-stage, initiation-interval-one pipeline: four tag comparators in each of 16 stages (64 comparator lanes across the pipeline), then eight control stages, with distributed stage-local tag reads, ideal tracker-state forwarding and a 64-entry label queue. The active L2 accepts at most one read per cycle, so its actual eligible callbacks respect this input rate. A single unpipelined four-comparator unit would not provide this throughput. Teacher actions become training-available at modeled completion. Parameter publication is coherent, and h/c state becomes available on scheduled inference completion. Pending work is serviced at scheduled simulator cycles even without a new demand callback. The conventional Stride comparator retains its original callback-time requests with no added modeled lookup delay. The finite sensitivity adds service costs to the neural inference and online update engines; it does not retrofit a Stride timing model. The zero-cost controls set modeled service/publication delays to zero while retaining actual CPU co-simulation work.

Design	Inference max KiB	Learning max KiB	Inference peaks KiB	Learning peaks KiB
h16 frozen ilm	NA	0.00	37.08	0.00
h8 frozen ilm	NA	0.00	17.58	0.00
h8 frozen ilm /64 /MAC0	16.95	0.00	14.95	0.00
h8 frozen ilm /64 /MAC16	16.95	0.00	15.68	0.00
h8 frozen ilm /64 /MAC4	16.95	0.00	15.12	0.00
h16 scratch s7	NA	2,722.53	37.12	515.28
h8 scratch s7	NA	1,625.84	17.58	397.72
h8 scratch s7 /64 /MAC0	16.95	1,625.84	14.95	397.72
h8 scratch s7 /64 /MAC16	16.95	1,633.30	15.59	338.62
h8 scratch s7 /64 /MAC4	16.95	1,633.30	14.93	341.10

Inference includes active weights, PC state, arithmetic scratch, active event/result payloads, and input/output queues. The learning subtotal adds the shadow teacher and its queue, examples, padded tensors, TBPTT activations, gradients, optimizer, training weights, publication buffer, and retained version bank. The zero-service unbounded rows have no finite inference maximum; measured occupied-state bytes remain in their peak envelope. The 64-entry rows provide the finite configured total. These are FP32 logical storage costs, not host RSS.

The category table groups runs with the same arm, hidden size, capacity and service case; its observed column takes the maximum across completed seeds/checkpoint budgets within that design. The following per-run totals keep runs separate.

Design	Storage category	Max (B)	Peak (B)
h16 frozen	active FP32 weights	20,880	20,880
h16 frozen	PC tags, valid/replacement metadata and h/c	NA	14,720
h16 frozen	inference scratch	2,176	2,176
h16 frozen	active event metadata	56	56
h16 frozen	decoded delta/address result payload	512	32
h16 frozen	input queue	896	56
h16 frozen	output queue	768	48
h16 frozen	shadow Stride / conventional Stride state	0	0
h16 frozen	teacher label availability queue	0	0
h8 frozen	active FP32 weights	7,632	7,632
h8 frozen	PC tags, valid/replacement metadata and h/c	NA	8,832
h8 frozen	inference scratch	1,344	1,344
h8 frozen	active event metadata	56	56
h8 frozen	decoded delta/address result payload	512	32
h8 frozen	input queue	896	56
h8 frozen	output queue	768	48
h8 frozen	shadow Stride / conventional Stride state	0	0
h8 frozen	teacher label availability queue	0	0
h8 frozen /64 /MAC0	active FP32 weights	7,632	7,632
h8 frozen /64 /MAC0	PC tags, valid/replacement metadata and h/c	6,144	6,144
h8 frozen /64 /MAC0	inference scratch	1,344	1,344
h8 frozen /64 /MAC0	active event metadata	56	56
h8 frozen /64 /MAC0	decoded delta/address result payload	512	32
h8 frozen /64 /MAC0	input queue	896	56
h8 frozen /64 /MAC0	output queue	768	48
h8 frozen /64 /MAC0	shadow Stride / conventional Stride state	0	0
h8 frozen /64 /MAC0	teacher label availability queue	0	0
h8 frozen /64 /MAC16	active FP32 weights	7,632	7,632
h8 frozen /64 /MAC16	PC tags, valid/replacement metadata and h/c	6,144	6,048
h8 frozen /64 /MAC16	inference scratch	1,344	1,344
h8 frozen /64 /MAC16	active event metadata	56	56
h8 frozen /64 /MAC16	decoded delta/address result payload	512	32
h8 frozen /64 /MAC16	input queue	896	896
h8 frozen /64 /MAC16	output queue	768	48
h8 frozen /64 /MAC16	shadow Stride / conventional Stride state	0	0
h8 frozen /64 /MAC16	teacher label availability queue	0	0
h8 frozen /64 /MAC4	active FP32 weights	7,632	7,632
h8 frozen /64 /MAC4	PC tags, valid/replacement metadata and h/c	6,144	5,472
h8 frozen /64 /MAC4	inference scratch	1,344	1,344
h8 frozen /64 /MAC4	active event metadata	56	56
h8 frozen /64 /MAC4	decoded delta/address result payload	512	32
h8 frozen /64 /MAC4	input queue	896	896
h8 frozen /64 /MAC4	output queue	768	48
h8 frozen /64 /MAC4	shadow Stride / conventional Stride state	0	0
h8 frozen /64 /MAC4	teacher label availability queue	0	0
No prefetch	active FP32 weights	0	0
No prefetch	PC tags, valid/replacement metadata and h/c	0	0
No prefetch	inference scratch	0	0
No prefetch	active event metadata	0	0
No prefetch	decoded delta/address result payload	0	0
No prefetch	input queue	0	0
No prefetch	output queue	0	0

Design	Storage category	Max (B)	Peak (B)
No prefetch	shadow Stride / conventional Stride state	0	0
No prefetch	teacher label availability queue	0	0
h16 scratch	active FP32 weights	20,880	20,880
h16 scratch	PC tags, valid/replacement metadata and h/c	NA	14,720
h16 scratch	inference scratch	2,176	2,176
h16 scratch	active event metadata	56	56
h16 scratch	decoded delta/address result payload	512	48
h16 scratch	input queue	896	56
h16 scratch	output queue	768	72
h16 scratch	shadow Stride / conventional Stride state	2,064	2,064
h16 scratch	teacher label availability queue	3,072	48
h16 scratch	bounded training examples and saved start states	23,552	22,528
h16 scratch	padded training input and target tensors	557,568	304,128
h16 scratch	TBPTT autograd-saved activations	2,097,152	94,416
h16 scratch	gradient tensors	20,880	20,880
h16 scratch	Adam moments and step tensors	41,824	41,824
h16 scratch	training FP32 weight copy	20,880	20,880
h16 scratch	publication buffer	20,880	20,880
h16 scratch	retained old inference weight bank	0	0
h8 scratch	active FP32 weights	7,632	7,632
h8 scratch	PC tags, valid/replacement metadata and h/c	NA	8,832
h8 scratch	inference scratch	1,344	1,344
h8 scratch	active event metadata	56	56
h8 scratch	decoded delta/address result payload	512	48
h8 scratch	input queue	896	56
h8 scratch	output queue	768	72
h8 scratch	shadow Stride / conventional Stride state	2,064	2,064
h8 scratch	teacher label availability queue	3,072	48
h8 scratch	bounded training examples and saved start states	15,360	14,336
h8 scratch	padded training input and target tensors	557,568	304,128
h8 scratch	TBPTT autograd-saved activations	1,048,576	48,464
h8 scratch	gradient tensors	7,632	7,632
h8 scratch	Adam moments and step tensors	15,328	15,328
h8 scratch	training FP32 weight copy	7,632	7,632
h8 scratch	publication buffer	7,632	7,632
h8 scratch	retained old inference weight bank	0	0
h8 scratch /64 /MAC0	active FP32 weights	7,632	7,632
h8 scratch /64 /MAC0	PC tags, valid/replacement metadata and h/c	6,144	6,144
h8 scratch /64 /MAC0	inference scratch	1,344	1,344
h8 scratch /64 /MAC0	active event metadata	56	56
h8 scratch /64 /MAC0	decoded delta/address result payload	512	32
h8 scratch /64 /MAC0	input queue	896	56
h8 scratch /64 /MAC0	output queue	768	48
h8 scratch /64 /MAC0	shadow Stride / conventional Stride state	2,064	2,064
h8 scratch /64 /MAC0	teacher label availability queue	3,072	48
h8 scratch /64 /MAC0	bounded training examples and saved start states	15,360	14,336
h8 scratch /64 /MAC0	padded training input and target tensors	557,568	304,128
h8 scratch /64 /MAC0	TBPTT autograd-saved activations	1,048,576	48,464
h8 scratch /64 /MAC0	gradient tensors	7,632	7,632
h8 scratch /64 /MAC0	Adam moments and step tensors	15,328	15,328
h8 scratch /64 /MAC0	training FP32 weight copy	7,632	7,632
h8 scratch /64 /MAC0	publication buffer	7,632	7,632
h8 scratch /64 /MAC0	retained old inference weight bank	0	0
h8 scratch /64 /MAC16	active FP32 weights	7,632	7,632
h8 scratch /64 /MAC16	PC tags, valid/replacement metadata and h/c	6,144	5,952
h8 scratch /64 /MAC16	inference scratch	1,344	1,344
h8 scratch /64 /MAC16	active event metadata	56	56
h8 scratch /64 /MAC16	decoded delta/address result payload	512	32
h8 scratch /64 /MAC16	input queue	896	896
h8 scratch /64 /MAC16	output queue	768	48
h8 scratch /64 /MAC16	shadow Stride / conventional Stride state	2,064	2,064
h8 scratch /64 /MAC16	teacher label availability queue	3,072	336
h8 scratch /64 /MAC16	bounded training examples and saved start states	15,360	14,240
h8 scratch /64 /MAC16	padded training input and target tensors	557,568	236,544
h8 scratch /64 /MAC16	TBPTT autograd-saved activations	1,048,576	47,712
h8 scratch /64 /MAC16	gradient tensors	7,632	7,632
h8 scratch /64 /MAC16	Adam moments and step tensors	15,328	15,328
h8 scratch /64 /MAC16	training FP32 weight copy	7,632	7,632
h8 scratch /64 /MAC16	publication buffer	7,632	7,632
h8 scratch /64 /MAC16	retained old inference weight bank	7,632	7,632
h8 scratch /64 /MAC4	active FP32 weights	7,632	7,632
h8 scratch /64 /MAC4	PC tags, valid/replacement metadata and h/c	6,144	5,280
h8 scratch /64 /MAC4	inference scratch	1,344	1,344
h8 scratch /64 /MAC4	active event metadata	56	56
h8 scratch /64 /MAC4	decoded delta/address result payload	512	32
h8 scratch /64 /MAC4	input queue	896	896
h8 scratch /64 /MAC4	output queue	768	48
h8 scratch /64 /MAC4	shadow Stride / conventional Stride state	2,064	2,064
h8 scratch /64 /MAC4	teacher label availability queue	3,072	336
h8 scratch /64 /MAC4	bounded training examples and saved start states	15,360	14,144
h8 scratch /64 /MAC4	padded training input and target tensors	557,568	247,104
h8 scratch /64 /MAC4	TBPTT autograd-saved activations	1,048,576	39,780
h8 scratch /64 /MAC4	gradient tensors	7,632	7,632
h8 scratch /64 /MAC4	Adam moments and step tensors	15,328	15,328

Design	Storage category	Max (B)	Peak (B)
h8 scratch /64 /MAC4	training FP32 weight copy	7,632	7,632
h8 scratch /64 /MAC4	publication buffer	7,632	7,632
h8 scratch /64 /MAC4	retained old inference weight bank	7,632	7,632
h16 warm ilm	active FP32 weights	20,880	20,880
h16 warm ilm	PC tags, valid/replacement metadata and h/c	NA	14,720
h16 warm ilm	inference scratch	2,176	2,176
h16 warm ilm	active event metadata	56	56
h16 warm ilm	decoded delta/address result payload	512	48
h16 warm ilm	input queue	896	56
h16 warm ilm	output queue	768	72
h16 warm ilm	shadow Stride / conventional Stride state	2,064	2,064
h16 warm ilm	teacher label availability queue	3,072	48
h16 warm ilm	bounded training examples and saved start states	23,552	22,528
h16 warm ilm	padded training input and target tensors	557,568	304,128
h16 warm ilm	TBPTT autograd-saved activations	2,097,152	94,416
h16 warm ilm	gradient tensors	20,880	20,880
h16 warm ilm	Adam moments and step tensors	41,824	41,824
h16 warm ilm	training FP32 weight copy	20,880	20,880
h16 warm ilm	publication buffer	20,880	20,880
h16 warm ilm	retained old inference weight bank	0	0
h8 warm ilm	active FP32 weights	7,632	7,632
h8 warm ilm	PC tags, valid/replacement metadata and h/c	NA	8,832
h8 warm ilm	inference scratch	1,344	1,344
h8 warm ilm	active event metadata	56	56
h8 warm ilm	decoded delta/address result payload	512	32
h8 warm ilm	input queue	896	56
h8 warm ilm	output queue	768	48
h8 warm ilm	shadow Stride / conventional Stride state	2,064	2,064
h8 warm ilm	teacher label availability queue	3,072	48
h8 warm ilm	bounded training examples and saved start states	15,360	14,336
h8 warm ilm	padded training input and target tensors	557,568	304,128
h8 warm ilm	TBPTT autograd-saved activations	1,048,576	48,464
h8 warm ilm	gradient tensors	7,632	7,632
h8 warm ilm	Adam moments and step tensors	15,328	15,328
h8 warm ilm	training FP32 weight copy	7,632	7,632
h8 warm ilm	publication buffer	7,632	7,632
h8 warm ilm	retained old inference weight bank	0	0
Stride	active FP32 weights	0	0
Stride	PC tags, valid/replacement metadata and h/c	0	0
Stride	inference scratch	0	0
Stride	active event metadata	0	0
Stride	decoded delta/address result payload	0	0
Stride	input queue	0	0
Stride	output queue	0	0
Stride	shadow Stride / conventional Stride state	2,064	2,064
Stride	teacher label availability queue	0	0

Run	Configured max KiB	Peaks/reservations KiB	Max / L2 bytes
h16 frozen ilm	NA	37.08	NA
h8 frozen ilm	NA	17.58	NA
h8 frozen ilm /64 /MAC0	16.95	14.95	0.07
h8 frozen ilm /64 /MAC16	16.95	15.68	0.07
h8 frozen ilm /64 /MAC4	16.95	15.12	0.07
h16 frozen i20m	NA	37.08	NA
h8 frozen i20m	NA	17.58	NA
No prefetch	0.00	0.00	0.00
h16 scratch s17	NA	552.40	NA
h16 scratch s27	NA	552.36	NA
h16 scratch s7	NA	552.40	NA
h8 scratch s17	NA	415.30	NA
h8 scratch s27	NA	415.34	NA
h8 scratch s7	NA	415.30	NA
h8 scratch s7 /64 /MAC0	1,642.79	412.67	6.42
h8 scratch s7 /64 /MAC16	1,650.24	354.21	6.45
h8 scratch s7 /64 /MAC4	1,650.24	356.03	6.45
h16 warm ilm s7	NA	552.40	NA
h8 warm ilm s7	NA	415.30	NA
Stride	2.02	2.02	0.01
h16 frozen ilm	NA	NA	NA
h8 frozen ilm	NA	NA	NA
h16 frozen i20m	NA	NA	NA
h8 frozen i20m	NA	NA	NA

A sum of component peaks and fixed workspace reservations is a conservative envelope; their maxima need not occur

simultaneously. Arithmetic scratch and active packet metadata are fixed reservations, not sampled allocator peaks. A configured total is NA if any component is unbounded or unreported. Per-category bytes/KiB and scope notes are retained in `storage.csv`.

Required predictor storage excludes measurement-only logging and host allocator/RSS overhead. Dedicated predictor SRAM is compared with the actual L2 capacity as a size ratio; it is not assumed to occupy L2 or directly pollute the cache. Configured bounds and observed peaks are reported separately.

Finite h8 design	MACs/cycle	IPC	Cycles (M)	Saved vs Stride (M)
h8 frozen ilm /64 /MAC0	0	0.40606	61.567	6.931
h8 frozen ilm /64 /MAC16	16	0.36290	68.889	-0.392
h8 frozen ilm /64 /MAC4	4	0.36254	68.958	-0.460
h8 scratch s7 /64 /MAC0	0	0.40560	61.637	6.860
h8 scratch s7 /64 /MAC16	16	0.36271	68.926	-0.428
h8 scratch s7 /64 /MAC4	4	0.36252	68.961	-0.463

Finite design	Eligible	Completed NN	Input drops	Training drops	Updates	State evictions
h8 frozen ilm /64 /MAC0	404,271	404,271	0	0	0	30
h8 frozen ilm /64 /MAC16	404,271	152,062	252,192	0	0	0
h8 frozen ilm /64 /MAC4	404,271	84,091	320,163	0	0	0
h8 scratch s7 /64 /MAC0	404,271	404,271	0	0	6,316	30
h8 scratch s7 /64 /MAC16	404,271	153,309	250,945	125,021	440	0
h8 scratch s7 /64 /MAC4	404,271	83,740	320,514	67,420	253	0

Finite online design	Milestone	First M	Confirm M	Updates at first
h8 scratch s7 /64 /MAC0	Saved cycles	1.0	3.0	253
h8 scratch s7 /64 /MAC0	99% finite frozen	2.0	4.0	505
h8 scratch s7 /64 /MAC16	Saved cycles	NOT REACHED	NA	NA
h8 scratch s7 /64 /MAC16	99% finite frozen	1.0	3.0	17
h8 scratch s7 /64 /MAC4	Saved cycles	NOT REACHED	NA	NA
h8 scratch s7 /64 /MAC4	99% finite frozen	1.0	3.0	9

h8 frozen ilm /64 /MAC0: IPC was 100.00% of its unbounded zero-cost control, with 290 additional end-to-end cycles. Completed inference covered 100.00% of eligible callbacks, with 0 input drops and 0 training drops. It finished with 0 queued inputs, 0 inference job, 0 pending training decisions, and 0 update in flight. This separates capacity loss from modeled service congestion.

h8 frozen ilm /64 /MAC16: IPC was 89.37% of its 64-entry zero-cost control, with 7,322,790 additional end-to-end cycles. Completed inference covered 37.61% of eligible callbacks, with 252,192 input drops and 0 training drops. It finished with 16 queued inputs, 1 inference job, 0 pending training decisions, and 0 update in flight. This separates capacity loss from modeled service congestion.

h8 frozen ilm /64 /MAC4: IPC was 89.28% of its 64-entry zero-cost control, with 7,391,367 additional end-to-end cycles. Completed inference covered 20.80% of eligible callbacks, with 320,163 input drops and 0 training drops. It finished with 16 queued inputs, 1 inference job, 0 pending training decisions, and 0 update in flight. This separates capacity loss from modeled service congestion.

h8 scratch s7 /64 /MAC0: IPC was 100.00% of its unbounded zero-cost control, with -208 additional end-to-end cycles. Completed inference covered 100.00% of eligible callbacks, with 0 input drops and 0 training drops. It finished with 0 queued inputs, 0 inference job, 47 pending training decisions, and 0 update in flight. This separates capacity loss from modeled service congestion.

h8 scratch s7 /64 /MAC16: IPC was 89.43% of its 64-entry zero-cost control, with 7,288,538 additional end-to-end cycles. Completed inference covered 37.92% of eligible callbacks, with 250,945 input drops and 125,021 training drops.

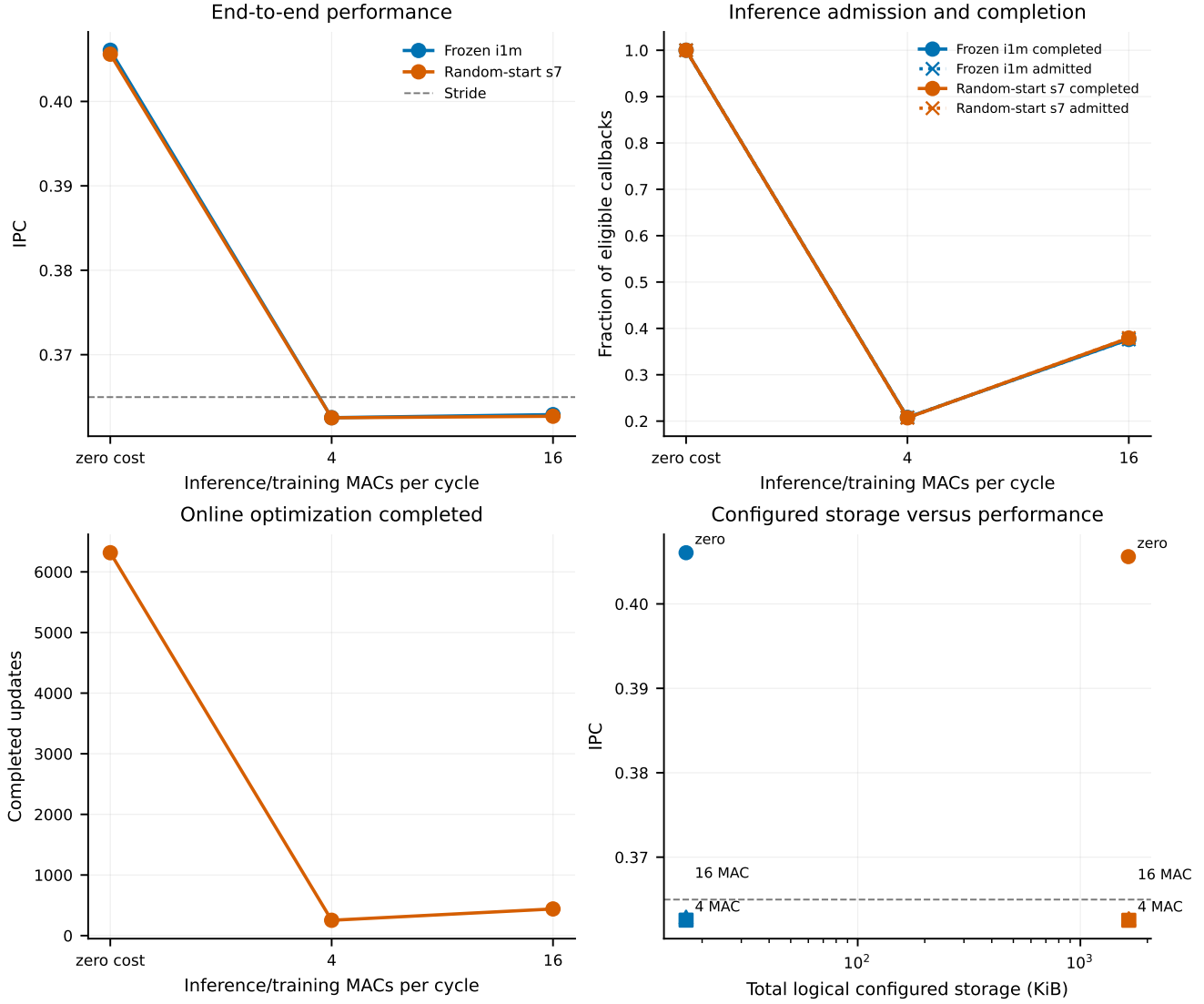


Figure 20: Measured h8 seed-7 sensitivity with the same 64-entry state table and bounded queues. Zero cost isolates finite capacity; 4 and 16 MACs/cycle add the predeclared inference, training and publication service model. Log storage axis compares configured logical FP32 storage, including the online trainer. Service costs are architectural assumptions, not measured silicon.

It finished with 16 queued inputs, 1 inference job, 64 pending training decisions, and 1 update in flight. This separates capacity loss from modeled service congestion.

h8 scratch s7 /64 /MAC4: IPC was 89.38% of its 64-entry zero-cost control, with 7,323,803 additional end-to-end cycles. Completed inference covered 20.71% of eligible callbacks, with 320,514 input drops and 67,420 training drops. It finished with 16 queued inputs, 1 inference job, 64 pending training decisions, and 1 update in flight. This separates capacity loss from modeled service congestion.

Finite design	MAC/infer.	Nonlinear/infer.	MAC/update	Nonlinear/update	Weight B/cycle
h8 frozen ilm /64 /MAC0	1,825.8	78.6	NA	NA	0.0
h8 frozen ilm /64 /MAC16	1,801.6	75.9	NA	NA	64.0
h8 frozen ilm /64 /MAC4	1,801.1	75.9	NA	NA	16.0
h8 scratch s7 /64 /MAC0	1,799.7	75.7	449,658.1	23,754.2	0.0
h8 scratch s7 /64 /MAC16	1,794.8	75.2	531,477.0	24,410.1	64.0
h8 scratch s7 /64 /MAC4	1,805.6	76.4	562,409.0	24,561.9	16.0

Across the completed timed arms, 97.85–98.24% of actual unique PQ enqueues resolved as redundant cache hits: the target line was already resident when the request was serviced. These are not late prefetches under the legacy L counter. Consequently legacy timeliness remained 98.87–99.92% while useful coverage was only 0.24–0.64%. The ratio $U/(U + L)$ excludes redundant-cache outcomes; a high value cannot establish useful coverage or a timing-costed win. The issue-cohort outcomes expose this distinction without redefining legacy timeliness.

Per-inference and per-update averages divide work by started jobs. MAC and service-cycle counters charge a full operation at its start, including a final operation that may extend beyond the retirement boundary; these totals are not measured engine utilization. Simulator completed updates and sample exposures advance at modeled training completion, while publication occurs later. A worker may finish an additional host batch during shutdown, so worker diagnostics do not replace the simulator completion/version counters. Raw MAC, nonlinear, state-traffic and publication-cycle totals, queue peaks, drops, and unfinished jobs are in `resources.csv`.

6.2 Measured host co-simulation costs

Run	Simulator wall s	Simulator RSS MiB	Worker CPU s	Worker RSS MiB
h16 scratch s17	253.01	10.99	131.65	281.98
h16 scratch s27	251.43	10.75	129.63	283.96
h16 scratch s7	251.38	10.79	130.40	282.27
h8 scratch s17	244.53	10.46	132.24	280.20
h8 scratch s27	228.15	10.55	114.71	280.23
h8 scratch s7	227.02	10.50	114.15	280.38
h8 scratch s7 /64 /MAC0	252.51	10.91	141.62	282.89
h8 scratch s7 /64 /MAC16	117.52	10.50	12.67	282.59
h8 scratch s7 /64 /MAC4	109.18	10.53	8.25	282.09
h16 warm ilm s7	253.18	10.66	131.47	282.31
h8 warm ilm s7	241.60	10.61	129.39	280.64

RSS includes whole host processes, libraries, allocators and IPC and is not predictor SRAM. Simulator wall time includes waits for the persistent CPU worker. Neither host wall time nor CPU time becomes simulated CPU stalls; congestion is accounted through the declared simulated engines and memory requests.

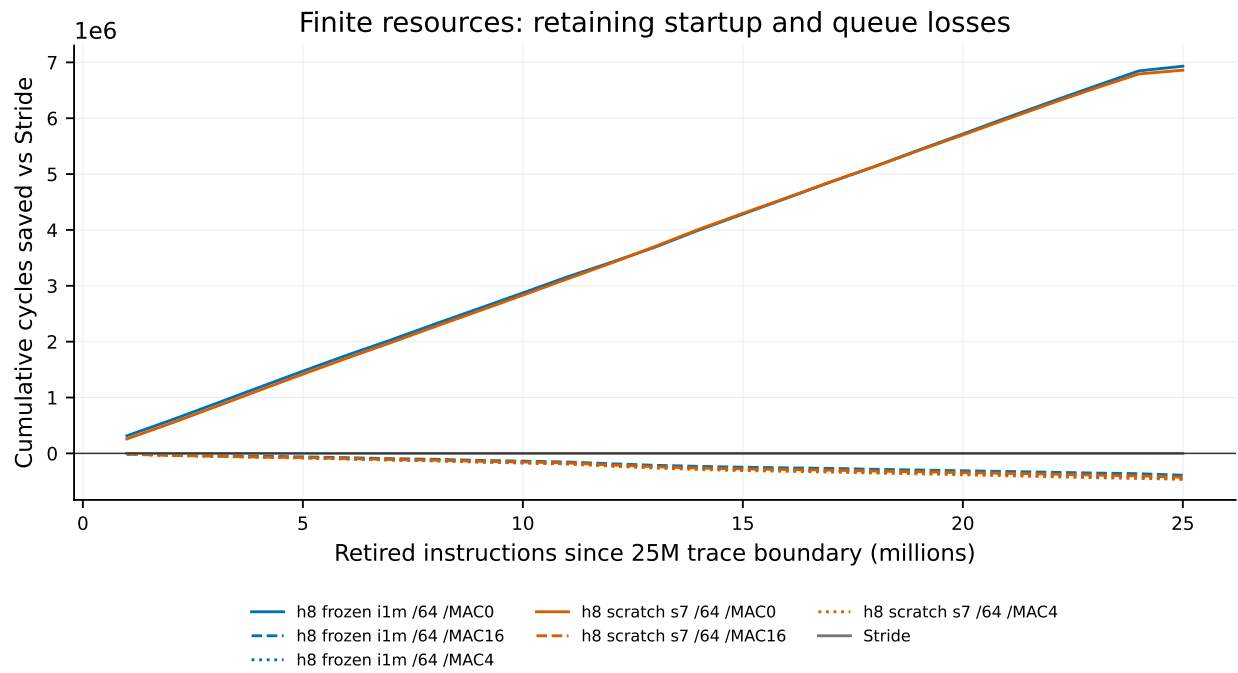


Figure 21: Each finite arm uses its own live closed-loop callback order. Solid, dotted and dashed lines denote zero cost, 4 MACs/cycle and 16 MACs/cycle respectively; color distinguishes frozen and scratch. The matched conventional Stride reference is shared only within the cold-start protocol.

7 Historical measurements: context, not replacements

The prior 1M plateau refers to the offline trace-start pretraining budget under a fixed 12-epoch Adam 0.002 recipe, with same-H comparisons to the i20m model. It does not mean an online learner became useful after one million observations. i1m supplied 16,172 eligible training decisions, 2,486 positive decisions and 4,875 action atoms; i20m supplied 325,389, 52,059 and 102,140. Offline updates numbered 228 and 4,404 respectively. Whole-trace PC grouping used there is not a causal online schedule when shared weights change.

Actual completed seed-7 live parity logs exist for all 28 exported prior points. From i1m through i20m, the maximum same-H relative IPC deviations were 0.21747% for h8 and 0.07318% for h16, meeting the earlier $\pm 0.5\%$ descriptive plateau rule. Request pressure and act rates still varied by roughly 13–15%; policy equivalence was not established. These warm-cache historical results are kept separate from new cold-start IPC.

The new historical compatibility runs match 120 of 120 original-parser fields exactly across 4 same-checkpoint comparisons. These comparisons include raw instruction/cycle/cache/prefetch counts and legacy derived ratios. Exact system-counter agreement does not by itself prove every internal floating-point value is identical; Python/C++ software parity is validated separately.

New compatibility model	Fields exact	Instructions	Cycles	Cycle difference
h16 i1m	30/30	25,000,003	61,014,571	0
h8 i1m	30/30	25,000,003	61,103,980	0
h16 i20m	30/30	25,000,003	60,983,624	0
h8 i20m	30/30	25,000,003	61,086,777	0

Historical model	Transport	IPC	Cycles	Coverage	Pressure
h8 i1m	offline keyed replay	0.40912	61,106,147	0.580	0.878
h8 i1m	frozen live parity	0.40914	61,103,980	0.580	0.952
h8 i20m	offline keyed replay	0.40924	61,088,223	0.576	1.016
h8 i20m	frozen live parity	0.40925	61,086,777	0.576	1.126
h16 i1m	offline keyed replay	0.40973	61,015,850	0.583	1.082
h16 i1m	frozen live parity	0.40974	61,014,571	0.583	1.209
h16 i20m	offline keyed replay	0.40994	60,984,797	0.585	0.974
h16 i20m	frozen live parity	0.40995	60,983,624	0.585	1.073

8 Limitations and interpretation

The experiment studies one held-out slice of one trace, two compact model capacities, three random-start seeds, and one pretrained seed. The recipe is fixed using the development gap before final evaluation. Progressive updates from already passed final events are allowed; later final measurements do not tune hyperparameters. Functional correctness with weak learning is a valid outcome.

Data scarcity is assessed using supervised observations and positive actions; optimization using actual updates/exposures and model behavior; state churn using evictions; traffic using requests, drops and cache outcomes; timing using queue/service/publication counters. A late-window improvement without cumulative recovery is reported as such. No composite score hides trade-offs, and absent milestones remain NOT REACHED.

The compact committed tables are `summary.csv`, `windows_1m.csv`, `windows_100k.csv`, `learning_milestones.csv`, `occupancy_milestones.csv`, `issue_cohorts.csv`, `storage.csv`, and figure inputs. Raw traces, checkpoints, logs, and training buffers remain outside tracked source. Build, test, pilot, run, status, resume, and report commands are in `command.md`.